Sensor structure shapes the format of cognitive maps in navigation agents

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Abstract

The format of spatial memory in the hippocampus varies systematically with a species' visual ecology, but whether this reflects a general sensor–memory coupling is hard to test directly: animal sensor ecology is fixed within species, and AI systems rarely co-train encoder and memory across a graded sensor design. We test this in silicon: a navigation agent where only the visual sensor varies along a graded scale of encoder spatial bandwidth, with architecture, task, and training held fixed. Across five conditions, agents reach near-identical task performance yet the hidden-state code for position differs in form. With a vision-limited encoder, position concentrates into a scene-invariant direction linearly readable from the hidden state; with a rich encoder, it distributes into non-linear directions and partly offloads to the encoder. Linear readability shifts faster than total position information — a format shift, not a total-information collapse. We frame this as an emergent *capacity allocation* between the encoder route and the memory-integration route, measured along three concomitant aspects (magnitude, format, consumption). Memory transplants are asymmetric (rich-encoder policies fail on bottleneck states; the reverse works), paralleling cross-modality remapping in hippocampal coding under hidden-state inference. Probe-readable position further dissociates from policy reliance: a bottleneck-encoder agent under-weights its readable code while a rich-encoder agent reads a non-spatial substitute. This positions sensor structure as a training-time lever on cognitive-map format, generates a falsifiable architecture-agnostic prediction for transformer navigators, and reframes bandwidth-restricting perception as potentially cognition-enabling rather than only compute-saving.

1 Introduction

Visual perception and downstream memory in artificial systems are commonly treated as a sequential pipeline, guided by the implicit assumption that scaling the encoder uniformly improves downstream representations. Recent benchmarks complicate this picture: vision–language models with strong visual recognition still fail at fine-grained spatial reasoning [Ramakrishnan et al., 2025, Yang et al., 2025]. Existing explanations have focused on the training distribution [Chen et al., 2024] or on differences in the geometric content of pretrained encoders [Tong et al., 2024, El Banani et al., 2024]. We instead question the underlying assumption: by treating perception and memory as a co-trained system, we ask how the structure of the encoder constrains what spatial information is accessible from memory through simple linear readout.

Cognitive neuroscience has long recognised that sensor structure shapes the format of cognitive maps. Across species, distinct sensory ecologies drive distinct spatial-coding strategies — foveated primates rely on view cells while near-panoramic rodents rely on place cells [Wirth et al., 2017, Rolls, 2024]; echolocating bats and congenitally blind humans show striking cross-modal remapping [Geva-Sagiv et al., 2016, Fortin et al., 2008]. Empirically isolating the causal role of sensor ecology is notoriously difficult in biology: body plans and evolutionary histories cannot be held

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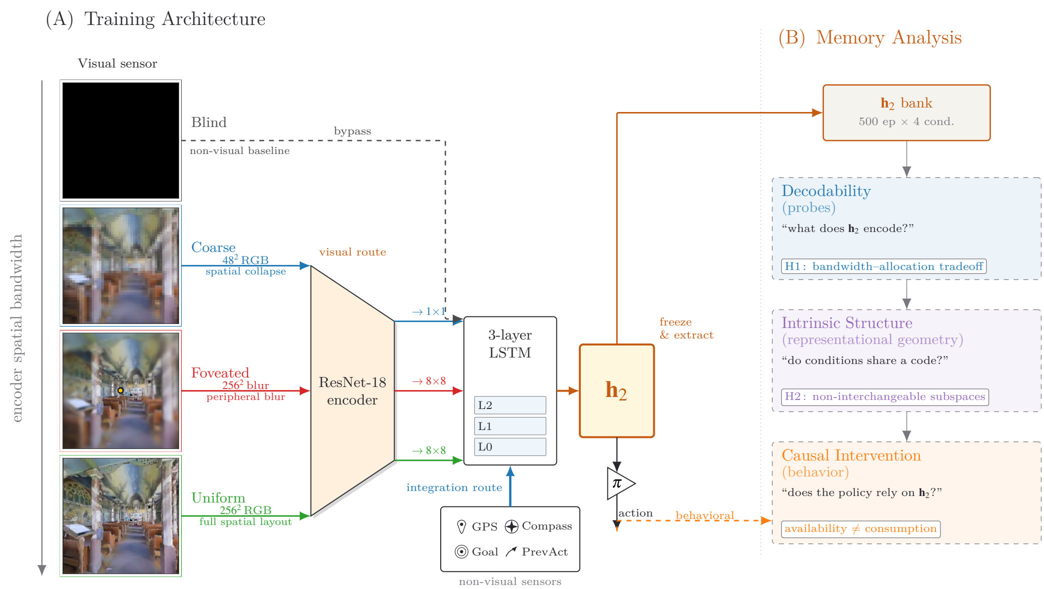


Figure 1: **Experimental setup and analysis pipeline.** *Row 1 (architecture):* four sensor conditions span a graded scale of encoder spatial bandwidth (*blind*, *coarse*, *foveated*, *uniform*; leftmost column), feeding a shared ResNet-18 encoder + 3-layer LSTM; *blind* bypasses the encoder. A log-polar foveation control (Appendix I) adds a 5th intermediate point on the same scale. *Row 2 (analysis):* three method blocks — decodability, intrinsic structure, causal intervention — read the frozen top-layer hidden state $\mathbf{h}_2 \in \mathbb{R}^{512}$ to test three views of one capacity allocation: *magnitude*, *format*, *consumption* (ğ2; synthesis in ğ4).

constant, so cross-species effects are confounded with everything else that differs between species. Artificial systems offer a powerful alternative, yet modern deep reinforcement learning rarely co-trains perception and memory across a systematically controlled sensor design.

We bridge this gap *in a silicon setting*. Building on deep-RL navigation agents known to develop cognitive-map-like representations even in the absence of vision [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018],[1] we introduce a comparative testbed that holds the encoder backbone, recurrent memory, task, optimisation, and reward entirely fixed, and varies a single degree of freedom: the spatial bandwidth of the visual sensor across five conditions (Figure 1).

We hypothesise an emergent *capacity-allocation* principle: the visual encoder and the recurrent integration of continuous sensor history (e.g., GPS) compete for a finite hidden-state budget. Our results confirm this tradeoff. As encoder spatial bandwidth increases, the linearly-readable, world-frame position signal in the recurrent state collapses monotonically while task performance stays near-perfect (93–99% success). We establish this reallocation along three concomitant views: *magnitude* (how much linearly-accessible position information lives in the recurrent state), *format* (which subspace carries it across conditions), and *consumption* (which route the policy actually uses for behaviour). Robustness against single-method artefacts is confirmed by convergent results across linear, non-linear (MLP), and information-theoretic (MINE) probes, leave-one-scene-out validation, and Procrustes shape distance. We further ground the format shift behaviourally with asymmetric memory transplants: rich-encoder policies fail to navigate using bottleneck states, demonstrating the representational change is mechanistically intrinsic and not an artefact of the probing framework [Orozco et al., 2025].

[1] "Cognitive map" in our paper is operationalised as the linearly-decodable world-frame position code at the policy-readable hidden state. Aspects of the broader concept [Epstein et al., 2017] are engaged by our LOSO scene-invariance test (relational across rooms) and predictive-horizon analysis (SR-style future-state code, Stachenfeld et al. 2017); hierarchical scene clustering and topological reachability are deferred to follow-up.

Our contributions are three folds. **(i) The capacity-allocation hypothesis:** a unifying framework explaining how visual bandwidth dictates spatial-memory format, organising five concomitant signatures (magnitude collapse, format divergence, consumption dissociation, scene-invariance, place-vs-view-cell character) into three concomitant views (magnitude, format, consumption). This matched-condition design enables an analytical avenue biology cannot easily realise: the systems-neuroscience toolkit applied to a *controlled comparative population* rather than a single recording. We treat this comparative methodology as an explicit secondary contribution, developed in full in ğ5. **(ii) Causal behavioural validation:** asymmetric memory transplants demonstrating that rich-encoder policies fail on bottleneck states, mechanically grounding the format shift and mirroring biological global remapping under cross-modality reorganisation. **(iii) Broader theoretical implications:** a falsifiable prediction for transformer-based navigators (ğ4) and an architecture-side reframing of bandwidth-restricting perception (foveation, log-polar) as cognition-*enabling* rather than only compute-saving — complementing data-side accounts [Chen et al., 2024] of VLM spatial-reasoning failures.

2 Methods

Architecture, task, and conditions (Figure 1, Row 1). PointGoal navigation in Habitat [Savva et al., 2019] trained with DD-PPO [Wijmans et al., 2020] across five conditions varying only the visual sensor: *blind* (no encoder); *coarse* ($48{\times}48$ RGB, $1{\times}1$ encoder feature map after ResNet-18 downsampling); *foveated* ($256{\times}256$ with eccentricity-dependent Gaussian blur $\sigma_{\max}{=}8$, $4{\times}4$ feature map); *uniform* ($256{\times}256$ unblurred, $4{\times}4$); *foveated-logpolar* ($\sim 2{\times}2$ via log-polar resampling onto a $64{\times}64$ retinal grid, control point isolating encoder-spatial-output from blur, Appendix I). All five share an identical 3-layer LSTM (512-d) and the Wijmans non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal); Gibson-0+ [Xia et al., 2018] and MP3D training pool; 18 MP3D test scenes for distribution-shift probes; 250M frames each; single seed per condition (cogsci modelling convention [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018, Schaeffer et al., 2023]; rigour from convergent multi-method evidence, ğ4.4); blind has an independently re-trained checkpoint (Appendix A). After training, weights are frozen and per-step top-layer hidden state $\mathbf{h}_2 \in \mathbb{R}^{512}$ is collected from $n = 500$ Gibson held-out deterministic-rollout episodes per condition.

Probing methodology (Figure 1, Row 2). Three method blocks read $\mathbf{h}_2$ to test three concomitant aspects of one capacity allocation:

- *Decodability → magnitude*: linear Ridge with Hewitt–Liang selectivity [Hewitt and Liang, 2019], 2-layer MLP, MINE [Belghazi et al., 2018], Skaggs spatial information [Skaggs et al., 1996]; plus step-binned, lag-k, predictive-horizon [Stachenfeld et al., 2017, Whittington et al., 2020], and excursion-forgetting variants for memory dynamics.
- *Intrinsic structure → format*: LOSO cross-validation [Sanders et al., 2020], cross-condition probe transfer, CKA [Kornblith et al., 2019], 1-NN purity, Procrustes shape distance [Williams et al., 2021], principal angles and position-direction cosines, participation ratio and TwoNN intrinsic dimension [Ansuini et al., 2019, Facco et al., 2017], eigenspectrum power-law [Stringer et al., 2019].
- *Causal intervention → consumption*: $5{\times}5$ memory transplant and memory-init transplant; paired same-scene shortcut-discovery (Tolman 1948 in silico); GPS-sensor mid-rollout ablation.

Each method is introduced at point of use in ğ3; full protocols and hyperparameters in Appendix B. Convergence across the three views on the same condition-level contrast is the strongest claim the methodology supports.

3 Results

The five agents reach near-identical task performance (Table 1) despite substantially different visual inputs — the encoder–memory–policy stacks must therefore internally compensate for the input differences, holding different internal representations of the same task. We read those representations along four views matching the three method blocks of Figure 1: how much linearly-readable posi-

Table 1: Per-condition headline. All five agents trained for 250M frames (ğ2); blind uses an independently retrained checkpoint (Appendix A). Probe R^2: 5-fold episode-level CV mean $\pm$ std on full-length trajectories (no per-episode step cap). Sub-chance R^2 in rich-encoder rows reflect long-tail collapse (mid-episode $R^2 \in [0.6, 0.95]$, Figure 10b); rich-encoder GPS R^2 is statistically indistinguishable from zero (one-sample t-test: foveated $p \approx 0.58$, uniform $p \approx 0.25$; df=4). Bold marks blind/coarse rows where GPS and compass remain decodable ($p < 0.001$) and best task performance.

Condition	SPL	Success	GPS R^2 (no-cap)	Compass R^2 (no-cap)
Blind	0.59	0.93	$+0.934 \pm 0.04$	$+0.844 \pm 0.03$
Coarse	0.853	0.989	+**0.582±0.099**	+**0.659±0.061**
Foveated	**0.894**	**0.993**	+0.178±0.659	+0.503±0.142
Foveated-logpolar	0.879	0.990	−0.029±0.759	+0.467±0.330
Uniform	0.891	**0.993**	−1.785±2.993	−1.244±3.579

tion lives in memory (magnitude, ğ3.1); what shape it takes spatially (ğ3.2) and temporally (ğ3.3); and what behaviour reads from memory under causal intervention (consumption, ğ3.4). Each view points to the same underlying capacity-allocation tradeoff between an encoder-derived visual route and a memory-integration route. The Discussion (ğ4) applies the same set of analyses to a different environment and encoder family, as a step toward a comparative cognitive neuroscience of artificial navigation agents.

3.1 Magnitude: linear position code shrinks with encoder bandwidth

Picture two routes by which the agent could know "where am I": integrating motion over time (works even without vision) or reading position-correlated visual features (only works with rich vision). Both can supply position to memory; the policy reads memory through a linear operation. As we widen the visual encoder, the agent reallocates from the first route to the second, and **the linearly-readable position signal at the policy-readable layer shrinks monotonically with encoder bandwidth** (Figure 2a). All five agents reach near-ceiling task performance (Table 1), so this is a representational dissociation, not a skill gap. The position information itself is not lost — a non-linear probe still recovers it (ğ3.2) — only the linear, policy-accessible component shrinks.

The bandwidth–magnitude relationship is robust against two natural alternative readings, and a sharper reading of the trigger emerges from the controls. The chance-level rich-encoder GPS R^2 is a real property of the hidden state rather than a limit of the probe: the same linear decoder reads distance-to-goal (DtG) at $R^2 \geq 0.81$ in every condition (Appendix C; Figure 18), so position-correlated structure is recoverable wherever it exists. The discriminator is not the number of encoder output cells either; FOVEATED-LOGPOLAR, whose encoder produces a $\sim 2{\times}2$ feature map close to coarse's $1{\times}1$, nonetheless lands in the rich-encoder regime — placing the relevant variable on per-step *feature variety*, the diversity of visual features delivered at each timestep, rather than cell count (Appendix I). Read this way, blind reproduces Wijmans et al. [2023]'s emergent-cognitive-map result for goal-driven agents without vision, and coarse extends it to a *vision-with-bottleneck* regime — a visual encoder is present, but its $1{\times}1$ output collapses per-step variety to one channel, leaving the same integration route to dominate the linear position code.

Capacity allocation between two routes. Memory has two routes to a position code — integrating the position sensor's signal over time, or reading position-correlated features from the encoder — and policy gradient biases capacity toward whichever route the encoder makes viable. Two views in the main figure are consistent. *Across training* (Figure 2b): sighted conditions start with high linear readability; rich-encoder conditions then decay along trajectories whose timescale tracks encoder informativeness, while coarse plateaus and blind stays stable. *Predictive horizon* (Figure 2c): the successor-representation cognitive-map signature [Stachenfeld et al., 2017] — memory predicting where the agent will be many steps ahead — is sustained only in blind, where the integration route carries the position code.

Corroborating diagnostics. Three appendix tests converge on the same picture (Appendix Figure 10). *(i) Pipeline view*: position readability is comparable across conditions where the position sensor first enters memory, and only diverges at the policy-readable layer — isolating the divergence to where the policy reads. *(ii) Peaked-unit content*: the most spatially-informative units in rich-encoder agents

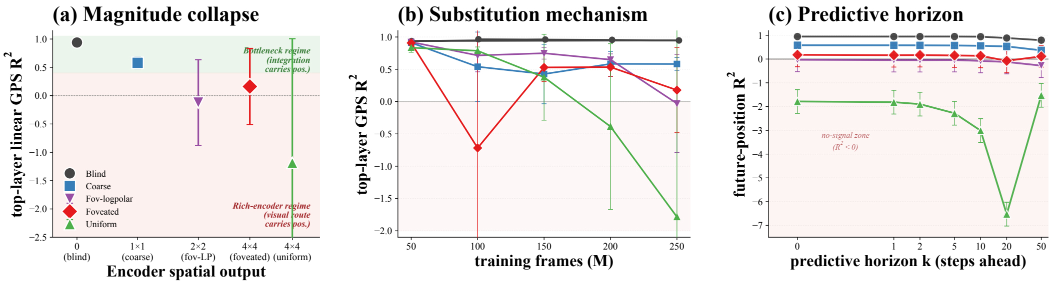


Figure 2: **Three views of magnitude.** *(a) Cross-condition:* top-layer linear GPS R^2 falls with encoder bandwidth. *(b) Across training:* rich-encoder conditions start high then lose readability; bottleneck conditions hold it throughout. *(c) Predictive horizon:* only bottleneck conditions linearly decode pos_{t+k} at long k. Shaded band: $R^2 < 0$, the linear probe is worse than predict-mean baseline — the "no-signal zone". Uniform's curve dives deep into this zone (its hidden state is scene-conditional, so 5-fold CV trains on some scenes' position-directions and tests on incompatible ones, producing systematically negative R^2); the dive is noise, not a meaningful trend.

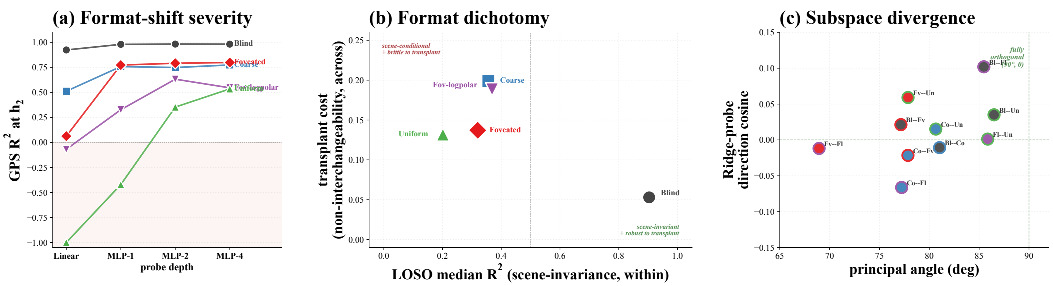


Figure 3: **Three views of spatial format.** *(a) Format-shift severity (probe-depth sweep):* GPS R^2 as a function of probe depth (linear $\to$ MLP-1 $\to$ MLP-2 $\to$ MLP-4) per condition. Bottleneck conditions plateau near $R^2 \approx 0.95$ from the very first column — linear already suffices, so deeper probes recover almost nothing more. Rich-encoder conditions ramp up sharply with probe depth — position is encoded but only non-linearly, and the depth needed to recover it grows with encoder bandwidth. *(b) Format dichotomy:* bottleneck sits in the scene-invariant + transplant-robust quadrant; rich-encoder conditions cluster in the scene-conditional + transplant-brittle quadrant. *(c) Subspace divergence:* all 10 condition pairs cluster near "fully orthogonal" (90ř, cos = 0), corroborating (b) at the geometric level.

track trajectory-phase and visual-landmark features rather than raw position, while in blind they track raw position — a view-cell-like vs. path-integrated dissociation [Skaggs et al., 1996, Rolls, 2024]. *(iii) Perturbation robustness*: a 25-step random-action mid-episode excursion fragments blind's position decoding most — the inverse of a passthrough memory and consistent at the high level with rodent path-integration under removed visual landmarks [Chen et al., 2016].

3.2 Spatial Format: collapsed linear readability is a format shift, not an information loss

Picture two ways the agent's memory might organise a position code: along a single direction shared across rooms (one "where am I" axis usable in any scene), or along scene-specific directions selected per context. The policy reads memory through a linear operation, so it can decode the first format but not the second. All five conditions *preserve* the position information itself — a non-linear probe recovers it where a linear probe is at chance, and information-theoretic estimates of total information between memory and position shift far less than the linear R^2 (Figure 3a; Appendix D.1) — but **the magnitude collapse of ğ3.1 is a *format shift*, not an information loss**: encoder bandwidth shifts which of the two formats the memory uses. The remainder of ğ3.2 characterises this shift along two complementary views: *within* a condition (does the readout direction depend on the scene?) and *across* conditions (do different conditions even share an axis?).

Within-condition: scene-invariance dichotomy. Leave-one-scene-out cross-validation gives a clean answer (Figure 3b, x-axis): bottleneck conditions generalise across held-out scenes (a single direction works in any room), while rich-encoder conditions do not (the position-direction differs per scene, so a single linear hyperplane cannot capture it; a non-linear probe can, by switching its readout on visual context — exactly the format-shift gap in Panel a). Intuitively, a bottlenecked encoder has no scene-specific features to bind to, so memory is forced toward a universal direction; a rich encoder lets memory exploit scene-specific features, fragmenting the position-direction across rooms. The dichotomy mirrors the no-remapping vs. global-remapping spectrum of hippocampal coding under hidden-state inference [Sanders et al., 2020, Achille and Soatto, 2018].

Across-condition: non-interchangeable subspaces. The picture sharpens further across conditions: each condition's position-direction lives in a mostly-orthogonal subspace of every other's (Figure 3c, all 10 pairs cluster near 90ř and direction-cosine ≈ 0) — the codes are not just *different*, they are nearly maximally distinct. A behavioural test confirms this in policy-execution terms (Figure 3b, y-axis): a 5×5 memory transplant shows a direction-asymmetry — a bottleneck-style memory state breaks rich-encoder policies, while a rich-encoder memory state still drives bottleneck policies, paralleling cross-modality remapping in echolocating bats [Geva-Sagiv et al., 2016]. Manifold-shape scalars (intrinsic dimension and eigenspectrum power-law α, Appendix Table 2) corroborate at the geometric level: blind separates on α (sharper variance fall-off, concentrated in a few directions) and on intrinsic dimension (lowest), with the four sighted conditions clustering on both metrics — the same bottleneck-vs-sighted contrast surfaced earlier in this section.

Synthesis. Spatial format is capacity allocation in geometric terms: bottleneck encoders push memory toward a scene-invariant single-direction code; rich encoders distribute capacity into scene-conditional directions occupying orthogonal subspaces. The first format is linearly readable by the policy; the second is not — this is the policy-side mechanism behind the magnitude collapse of ğ3.1. The structural separation is laid down well before convergence (Appendix Figure 15c).

3.3 Temporal Format: blind recomputes the code each step, sighted read it as stationary

Memory in a navigation agent can take two shapes. *Stationary code:* fix a readout axis at the start of the episode and let its projected value update as the agent moves — coordinate meanings stay put. *Rotating code:* rotate the readout axis step by step as new motion is integrated — every coordinate's meaning shifts. **Blind takes the rotating path; the four sighted conditions take the stationary path; within sighted, the less encoder bandwidth a condition has, the stiffer (more stationary) its readout becomes.** The mechanistic story: when vision delivers fewer high-frequency scene updates, memory has to act as the scene anchor, and anchoring works better with a stable readout direction.

Step-by-step decoder generalisation. Figure 4(a): train a linear decoder of egocentric goal-vector at trajectory step t_{train} and evaluate it at every other step t_{test} [King and Dehaene, 2014]. A stationary readout produces a square block (the same decoder works everywhere); a rotating readout produces a diagonal stripe (the decoder only works near its training step). Blind is pure diagonal (no transfer at lag ± 30); coarse is the largest red block (the most stationary readout); fov-LP, foveated, and uniform are graded intermediates. Our pre-registered prediction had blind as the stationary one; we report what we found.

Two quantitative views on one matched episode. Place all five agents at the same start with the same goal in the same Gibson scene (Almena, episode 157). All five complete it. *Figure 4(b) — total memory work.* Cumulative L2 path length of $\mathbf{h}_2$ in the 512-d recurrent state space, integrated step by step. Blind's curve ends at 1453 over 350 winding steps; the four sighted conditions stop early at 309–467 over 79–101 direct steps (coarse $467 >$ fov-LP $435 >$ uniform $330 >$ foveated 309). *Figure 4(c) — format stationarity.* Mean decoder R^2 as a function of trajectory-step lag $|i - j|$, with each panel-(a) TGM heatmap collapsed to a single curve. **The five curves fan out monotonically: blind crashes to ≈ -1.2 within the first few steps (catastrophic transfer, the diagonal-only signature); coarse holds $R^2 \approx +0.18$ at small lag (most stationary block); fov-LP $+0.13$, foveated $+0.13$, uniform $+0.08$.** Panel-(c)'s ranking exactly inverts encoder bandwidth (§3.2): less bandwidth, more stationary readout. Panel (b) agrees on the broad strokes (coarse at the stationary-and-busy extreme, blind off-scale) but reorders fov-LP, foveated, uniform slightly — panels (b) and (c) measure different things (*how much* memory moves vs. *whether the readout axis stays put*), and a stable axis can carry a changing projected value. Together: blind's memory is a *slow-rotating integrator* reading off a

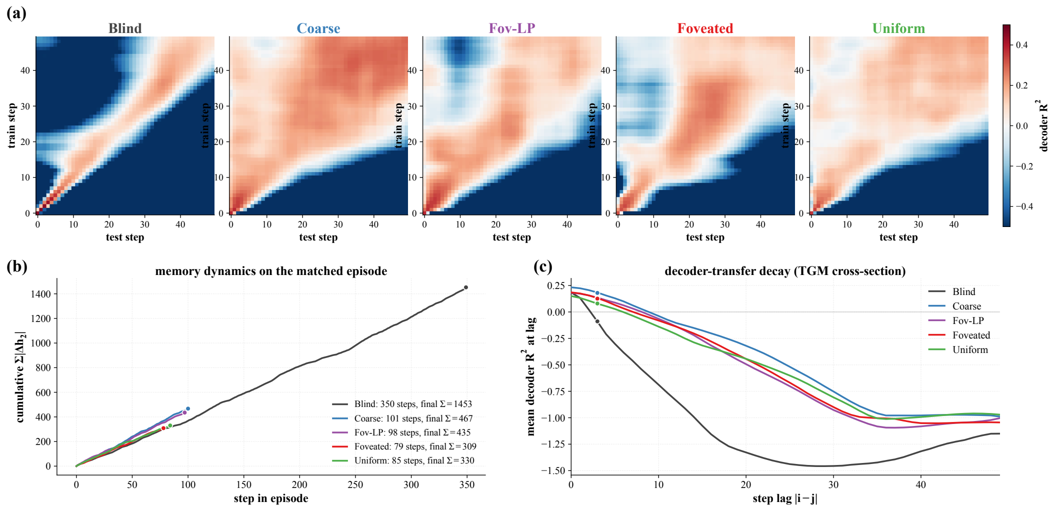


Figure 4: **Temporal format: blind rotates the readout axis step by step; sighted hold it stationary, with stiffness graded by encoder bandwidth. (a)** Step-by-step decoder generalisation [King and Dehaene, 2014]: each $50{\times}50$ heatmap is decoder R^2 trained at row step, tested at column step. Diagonal-only $\Rightarrow$ rotating basis (blind); square block $\Rightarrow$ stationary readout (coarse); intermediates $\Rightarrow$ fov-LP, foveated, uniform. **(b, c)** One matched episode, identical start/goal across all five conditions: **(b)** cumulative memory displacement $\Sigma|\Delta\mathbf{h}_2|$ vs step-in-episode (total memory work); **(c)** mean decoder R^2 vs step lag $|i-j|$ (format stationarity, each panel-(a) TGM collapsed to a curve). Panel (c) ranks coarse $+0.18 >$ fov-LP/foveated $+0.13 >$ uniform $+0.08$, exactly inverting encoder bandwidth. Murray τ and persistent-homology corroborations in App F.

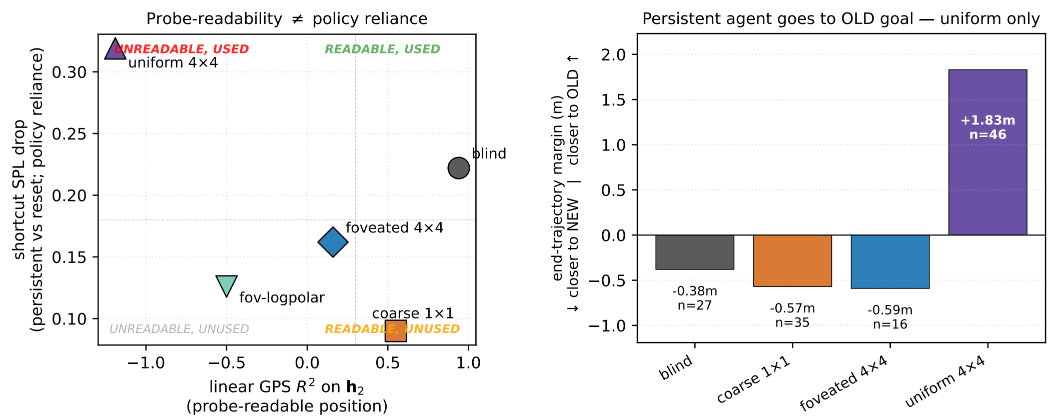
step-by-step rotating basis; sighted memory is a *stationary readout* whose stiffness scales inversely with encoder bandwidth. Murray-style per-unit autocorrelation ($\tau \approx 12$ blind, $\tau \approx 7$–8 sighted) and a persistent-homology corroboration [Gardner et al., 2022] are in Appendix F.

3.4 Consumption: causal intervention dissociates probe-readability from policy reliance

A linear probe tells us *where in memory the position code lives*, but not whether the policy actually uses it. The two could come apart in either direction: a condition might carry a probe-readable position code that the policy ignores ("readable but unused"), or might lack one yet behave as if it had one ("unused but readable"). Both cases would warn against reading probe results as direct policy mechanism. We test policy reliance with a paired same-scene different-goal shortcut protocol [Tolman, 1948]: re-init the recurrent memory between episodes vs. persist it; the SPL drop measures how much the policy depends on persistent memory. **Probe-readability and policy reliance dissociate in two off-diagonal cases.**

Coarse carries a linear position code yet has a small shortcut SPL drop ("readable but unused" — the policy under-weights its own readable code); uniform has no linear position code yet has the largest shortcut SPL drop ("unused but readable via a substitute" — the policy reads something else from memory) (Figure 5a). A direct readout test names the substitute: a scene-id classifier separates scenes substantially better in uniform than in blind, while within-scene linear position decoding goes the other way — uniform's memory state appears to encode scene identity more strongly than position, the inverse of blind. Trajectory-level analysis (Figures 5b, 25) gives the candidate mechanism: only uniform's persistent agent ends up closer to the *previous* episode's goal than the new one — a visual / scene-history anchor ("head to where you were going last episode") is one consistent reading.

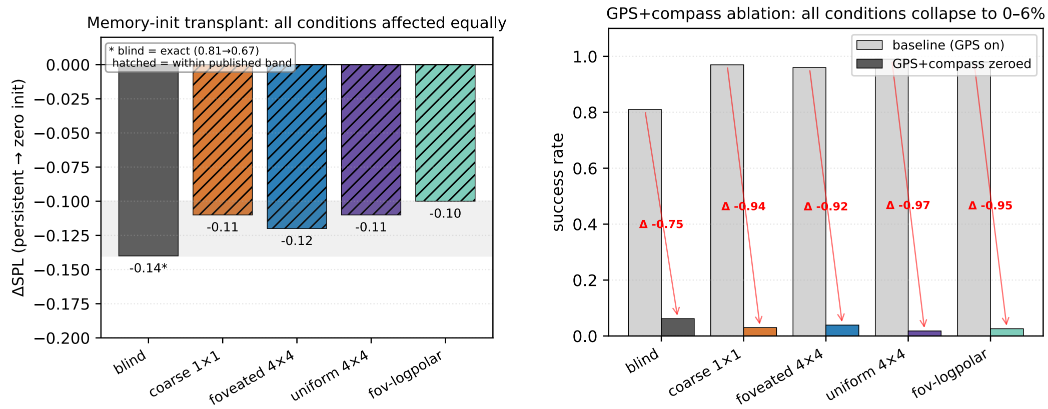
Boundary checks: upstream and downstream causal interventions. Two further causal stress-tests characterise the consumption mechanism without separating conditions (Figure 6). *Memory-init transplant* (re-initialise the agent's memory from the previous-episode end-state rather than from zero): all five conditions show a comparable SPL drop (Figure 6a). The cross-condition uniformity is itself the finding — every policy at $t{=}0$ expects a zero memory state, regardless of encoder. **Memory**



(a) Probe-readability $\neq$ policy reliance **(b) Uniform persistent agent goes to OLD goal**

Figure 5: **Probe-readable position is not what drives behaviour.** *(a) Off-diagonal cases:* per-condition shortcut SPL drop (y-axis = policy reliance on persistent memory) vs. linear position decoding from memory (x-axis = probe-readability). Coarse sits in "readable but unused", uniform in "unused but readable via a substitute". *(b) The substitute uniform reads:* end-trajectory margin (distance from persistent end to OLD goal − distance to NEW goal) on same-floor failure cases. Only uniform ends closer to the previous episode's goal — consistent with reading a scene-history anchor instead of the absent position code. Foveated-logpolar is excluded from (b): it has the most robust persistent-memory behaviour of the five conditions and therefore yields too few paired-failure cases (n=2) to estimate a margin.

in this architecture is a dynamics-only *activity-slot* store, not a synaptic store [Dorrell et al., 2024]: persistent memory is bound to the trajectory dynamics, not to the underlying weights. *GPS-sensor ablation* (zero GPS+compass from step 0) collapses all five conditions to near-zero success (Figure 6b): the position sensor is causally critical across the spectrum. The absolute drops cluster tightly across the four sighted conditions and do not cleanly track the magnitude vs. consumption split; we read this as a check on the upstream sensor's causal role rather than independent ranking-level evidence for the integration-vs-substitute split established above. Boundary tests, classifier protocol, and the failed mid-rollout visual ablation are detailed in Appendix B.



(a) Memory-init transplant **(b) GPS+compass ablation**

Figure 6: **Causal interventions: memory is dynamics-bound, and the position sensor is causally critical across all conditions.** *(a) Memory-init transplant:* re-initialising memory from the previous episode's end-state (instead of zero) drops SPL by a comparable amount across all five conditions. The cross-condition uniformity is the finding — every agent's policy expects a zero memory state at $t=0$, so memory acts as an activity-slot store bound to the trajectory dynamics. *(b) GPS+compass ablation:* zeroing GPS and compass from step 0 collapses success to near-zero across all conditions. The drops cluster tightly across the four sighted conditions; blind drops less because its baseline is already lower. The position sensor is the upstream causal source whose integration the memory encodes.

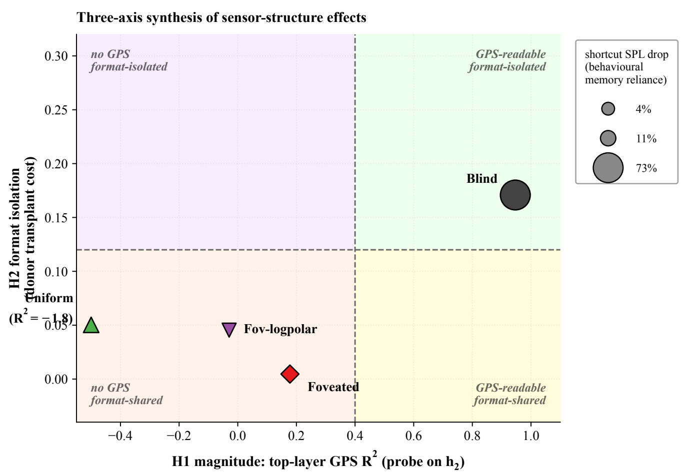


Figure 7: Synthesis: three views of one capacity allocation. *X-axis:* top-layer GPS R^2 on $\mathbf{h}_2$ (magnitude). *Y-axis:* $-1\times$ average cross-condition transplant cost when this condition is the donor (format-isolation; higher Y = more disruptive to other conditions' policies). *Marker size:* second-episode shortcut SPL drop $\%$ (consumption; larger = more policy dependence on persistent recurrent memory). Within each axis, multiple convergent measurements — e.g. information-allocation, scene-invariance, predictive-horizon, place-cell, and excursion-fragility findings (Figures 3, 10) — sharpen the *interpretation* but are not separate views of variation.

4 Discussion

4.1 Synthesis: three views of one capacity allocation

A reader could imagine three independent things going on across our five conditions — they differ in how much position lives in memory, in how that position is encoded, and in whether the policy uses it. We argue the three are views of one allocation: the visual encoder and the integration of the position sensor compete for finite memory capacity, and the encoder's bandwidth tilts the balance. **Bottleneck conditions retain a linear position code in memory, rich-encoder conditions reallocate to the visual route, and the policy reads from whichever route the memory makes available.** Each condition occupies a distinct (X, Y, size) coordinate in Figure 7: magnitude on X (linear position decode, ğ3.1), format-isolation on Y (cross-condition transplant cost, ğ3.4), policy reliance as marker size (shortcut SPL drop, ğ3.4). Foveation occupies a hybrid position: it shares uniform's rich-encoder magnitude but separates on every other measure — most strikingly, foveated agents recover a held-out-MP3D position code that uniform agents do not, consistent with the visual and integration routes being independent capacity sinks rather than two ends of one knob. Coarse and uniform are both on the same magnitude side of the diagram for different reasons: coarse's encoder cannot supply spatially-resolved features (so capacity defaults to integration), while uniform's encoder supplies rich features (so capacity is reallocated to the visual route, leaving integration unattended). The two are different regimes of one allocation, not the same mechanism.

4.2 Biological precedent and predictions

The hypothesis "sensor niche shapes spatial memory" has long-standing comparative-cognition precedent — rodent dark-rearing [Chen et al., 2016], bat cross-modality remapping [Geva-Sagiv et al., 2016], congenitally blind humans with larger anterior hippocampal volume [Fortin et al., 2008]. We do not claim that our silicon results *demonstrate* the bio-AI parallel; we offer them as patterns from controlled within-architecture variation that one would expect if the hypothesis held in animals. Loosely mapped: blind $\leftrightarrow$ subterranean rodents [Kimchi et al., 2004]; coarse $\leftrightarrow$ single-channel sensory summaries (echolocation, infrared pit-detection); foveated $\leftrightarrow$ primate / raptor (gaze-fixed); uniform $\leftrightarrow$ insect compound eye / zebrafish near-isotropic sampling [Geva-Sagiv et al., 2015, Toledo et al., 2020] — interpretive scaffolding, not mechanistic identity claims.

Two specific bio-AI bridges land cleanly under the framework of Sanders et al. [2020], in which place fields remap when sensory evidence supports distinct hidden states. (i) Our LOSO catastrophe in rich-encoder conditions matches global-remapping (rich per-step visual evidence supports scene-specific hidden states; the linear position direction reorganises across rooms); bottleneck conditions exhibit no remapping (a single scene-invariant code). (ii) Blind's transient-code temporal-generalisation signature and longer intrinsic timescale [King and Dehaene, 2014, Murray et al., 2014] reproduce the cortical-hierarchy slow-integrator pole. Both are "the same metric one would run on monkey or human cortex, applied unchanged to a recurrent navigator". Three falsifiable predictions follow: *(i)* within-species manipulations of effective per-step visual variety (gaze restriction, dark-rearing, prosthetic compound vision) should shift hippocampal coding along the no-/global-remapping spectrum; *(ii)* gaze-fixed primates should show coding closer to rodent place cells than to free-gaze primate view cells, with Wirth et al. [2017] the closest existing dataset; *(iii)* the Skaggs, LOSO, predictive-horizon, MINE, and subspace-divergence findings, catalogued separately in cognitive map theory [Tolman, 1948, O'Keefe and Nadel, 1978, Stachenfeld et al., 2017, Whittington et al., 2020], may all be views of a single mechanism. Whether biological hippocampi obey a finite-capacity allocation principle remains an open empirical question.

4.3 Implications and scope

The principle has a simple geometric reading. A finite memory has a fixed number of dimensions to spend; how it spends them depends on what its input already provides. When the encoder delivers rich, near-complete spatial information at every step, the memory has no reason to maintain a separate linearly-readable position code, and the policy is free to commit those dimensions to other content. When the encoder delivers almost nothing, the memory must reconstruct position by integrating the action stream over time, and the resulting code is forced into a form the linear action head can consume. **This is the empirical pattern of ğ3.1–ğ3.2, and it is what the invariance–disentanglement reading of the information bottleneck [Tishby et al., 1999, Achille and Soatto, 2018] predicts:** bandwidth-restricted memory is the architectural mechanism that drives the trained representation toward scene-invariant rather than scene-conditional spatial encoding (Figure 3b). The framing has a quantitative skeleton: a hidden state with effective rank d allocates dimensions among (i) directly carrying encoder-derived features (capacity d_e), (ii) integrating sensor history (capacity d_i), and (iii) carrying task-relevant non-position content (capacity d_o), subject to $d_e + d_i + d_o \lesssim d$. Bottleneck encoders force $d_e \to 0$, leaving d_i to dominate the linear position code; rich encoders enlarge d_e, so policy gradient minimises d_i down to whatever residual the integration route still needs. Our Procrustes / participation-ratio / eigenspectrum-α measurements collectively probe d (effective rank ≈ 4 for blind, lower for coarse where the encoder collapse shrinks d overall, Appendix E.3); the magnitude collapse follows from d_i shrinking as d_e grows. We use IB descriptively rather than mechanistically; Saxe et al. [2019]'s critique of IB-deep-learning theory cautions against absolute MI-magnitude claims, but the cross-condition *ordering* reported here (linear collapse, non-linear-probe recovery, mutual-information rank in Appendix D.1) is robust to those concerns.

The principle is also architecture-agnostic: a content-level allocation should appear in any system where a finite-capacity memory reads a pretrained encoder. Replacing recurrent gating with attention-over-history should not change the headline ordering. In transformer-based navigators (e.g. Zeng et al. 2024; cf. Whittington et al. 2022 for the transformer–hippocampus correspondence), where "memory" lives in the KV cache or attention over rollout history, the principle predicts the same monotonic anti-correlation reported in Figure 2a: **attention-readable position codes under bandwidth-restricted encoders, attention-internal position codes only non-linearly decodable under rich encoders**. The cleanest falsification target is direct — train a transformer navigator under our five sensor conditions; the opposite pattern would falsify architecture-independence and bound the principle to recurrent-state systems. The same logic re-frames recent VLM spatial-reasoning failures [Ramakrishnan et al., 2025]: rich pretrained encoders may divert capacity *away from* the downstream substrate's spatial code, rather than merely fail to teach it, and bandwidth-restricting perception (foveation, log-polar) may be *cognition-enabling* rather than only compute-saving. The relevant variable is encoder spatial-feature *variety* per step rather than raw input bandwidth (Appendix I); a finer multi-point sweep within recurrent agents is left to future work.

As a partial test of architecture-independence, we ported the encoder-bandwidth design to a different environment, encoder family, and training regime (Figure 8). Frozen DINOv2-Base [Oquab et al., 2024] patch features at five resolutions — mirroring our Habitat chassis (BLIND, COARSE, FOVEATED, UNIFORM, FOVEATED-LOGPOLAR) — feed a small recurrent memory trained on Memory Maze [Pasukonis et al., 2023] 9×9 offline trajectories with next-feature prediction. We then probe `agent_pos` from the memory state (Ridge linear and 4-layer non-linear probe, frame-level CV with 80/20 random split, eval window steps 250–500). Three pre-registered hypotheses land cleanly. *(1) Linear R^2 rises monotonically with encoder bandwidth:* BLIND $0.01 \rightarrow$ COARSE $0.38 \rightarrow$ FOVEATED $0.41 \rightarrow$ FOVEATED-LOGPOLAR $0.41 \rightarrow$ UNIFORM 0.45. *(2) Non-linear R^2 saturates near* 0.998 *for all sighted conditions* and falls to chance for BLIND (-0.0003). *(3) The non-linear gap (non-linear − linear) decreases monotonically with bandwidth:* COARSE $+0.62$, FOVEATED $+0.59$, FOVEATED-LOGPOLAR $+0.59$, UNIFORM $+0.55$. **Format-shift recovery is largest where linear readability is smallest — the same dichotomy our Habitat result reports, in a different environment, encoder family, and policy regime.** *Direction relative to Habitat.* The Memory-Maze setup deliberately omits the position-sensor channel our Habitat agents receive; without that explicit positional anchor, the encoder is the only sensory route to absolute position, and BLIND cannot decode it (no signal, no integration anchor). The bandwidth-allocation *tradeoff* (where blind's integration route compensates) thus requires a competing route; the format-shift dynamic (linear-readable code emerges from a non-linear substrate as bandwidth grows) generalises across both regimes. The two together place the broad "encoder structure shapes memory format" claim on a different environment, encoder family, and policy regime, while clarifying that the specific direction (high-bandwidth-rich vs. bottleneck-rich) is mediated by which sensory channels carry positional information.

4.4 Limitations

(i) Single seed per condition. Following the comparative-cognition modelling convention [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018, Schaeffer et al., 2023]. Cross-condition rigour comes from convergent multi-method evidence: linear and non-linear probes, CKA, Procrustes, subspace divergence, LOSO, lag-k, 5×5 memory transplant, Skaggs spatial information, and excursion-forgetting all rank the conditions in the same direction — unlikely under single-seed noise alone — supplemented by an independent-retraining replication of the blind condition (Appendix A). Per-condition seed-variance error bars over a multi-seed grid are left for follow-up.

(ii) Method-family scope. Methods we do not report and that could change a finding's strength: dynamical-systems analysis (fixed-point / slow-point structure [Sussillo and Barak, 2013]; Dynam-

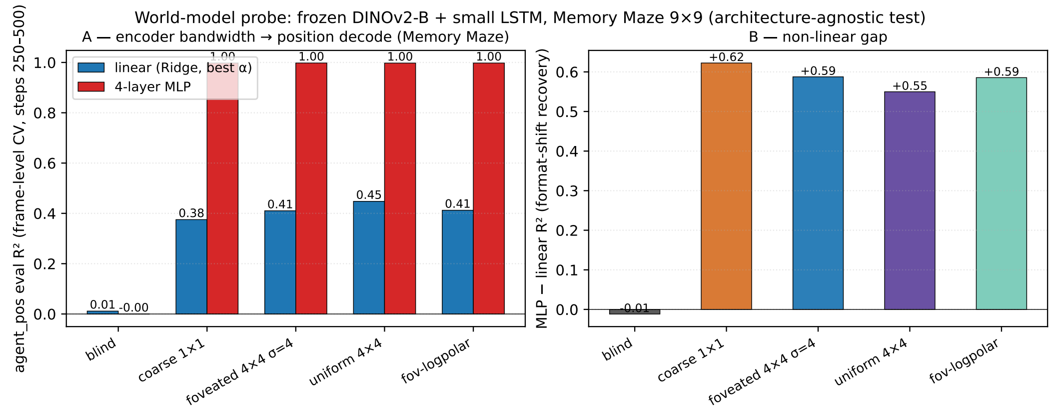


Figure 8: **Architecture-agnostic confirmation: format shift reproduces on Memory Maze with a frozen DINOv2-Base encoder feeding a small LSTM.** *(A)* Linear (Ridge, best of $\alpha \in \{0.1, 1, 10, 100, 1000\}$, blue) and 4-layer MLP (red) eval R^2 for `agent_pos` from the LSTM hidden state, frame-level random CV (80/20), eval window steps 250–500 of T=500 Memory-Maze trajectories. Linear R^2 is monotone-increasing with encoder bandwidth (BLIND $0.01 \rightarrow$ UNIFORM 0.45); MLP R^2 saturates at ≈ 0.998 for all sighted conditions and is at chance for BLIND. *(B)* The non-linear gap (MLP−linear) is monotone-*decreasing* with bandwidth: format-shift recovery is largest where linear readability is smallest (COARSE $+0.62$ vs. UNIFORM $+0.55$), the same dichotomy our Habitat agent shows in Figure 3a. The opposite linear-R^2 direction relative to Habitat reflects the absence of GPS/compass channels in the Memory-Maze setup — without an alternative positional anchor, the encoder is the only route, and BLIND carries no signal.

ical Similarity Analysis [Ostrow et al., 2023] as a dynamics-aware Procrustes alternative) would speak to format at the level of computational primitives. We piloted DSA on our memory-state time-series (10-d PCA pre-reduction, 50-step trajectories) and found the cross-condition signal sensitive to hyperparameter choices (`n_delays`, rank), rollout protocol (excursion-warmup vs. deterministic-rollout), and episode-phase alignment (blind has median episode length $\sim 2.5\times$ sighted, so first-T-step samples occupy different behavioural phases). Under the cleanest protocol (deterministic-rollout, all conditions, T=50), sighted clustered at floor ~ 0.10 angular distance while blind sat further out (~ 0.27); but blind's within-condition split-half DSA was also ~ 0.23, leaving the signal-to-noise ratio insufficient to publish a definitive ranking. Information-theoretic estimators of $I(h_t; \mathrm{pos}_{t+k})$ would replace decodability R^2 with a nat-scale predictive-horizon measure; gaze-placement controls and a $\sigma_{\max}$ strength sweep within the foveated condition would test the rich-encoder content. All four are left for follow-up.

(iii) Architecture, algorithm, and task scope. We use a recurrent (LSTM) backbone with DD-PPO, PointGoal, and a ResNet-18 visual encoder, matching the canonical pipeline of [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018] for direct comparability; the architecture choice is replication-driven rather than central to the claim, and the principle is stated at the encoder–memory interface (architecture-agnostic). A reasonable concern is whether the encoder–memory race is a recurrent-gating artefact (e.g. vanishing-gradient limitations specific to LSTMs); two within-paper checks argue against it: *(a)* non-linear (2-layer MLP) probes recover position in rich-encoder conditions ($R^2 \in [0.35, 0.79]$), so the memory *does not* discard position — only the linear-readable form tracks encoder bandwidth, a property of the linear policy head; *(b)* the cross-condition memory transplant (ğ3.4) is a behavioural test outside any probe framework and reproduces the same ordering. The encoder–memory race is in principle architecture-agnostic; the cleanest falsification target is a transformer-based navigator under the same conditions, identified in ğ4 but not verified here. The magnitude and format patterns could also shift under different RL algorithms, different navigation tasks (ObjectGoal, ImageGoal, exploration), or different visual encoders.

5 Conclusion

A five-condition visual-sensor ablation in a recurrent PointNav agent shows sensor structure as a representational-content variable whose effects cohere under a single capacity-allocation principle measured as three views: *magnitude* (bandwidth–allocation tradeoff between encoder and recurrent-integration routes), *format* (within-condition scene-invariance, across-condition non-interchangeable subspaces with asymmetric memory transplant, and a King–Dehaene-style temporal-format dissociation in which the blind code rotates over time while the coarse code stabilises), and *consumption* (probe-readable position dissociates from policy reliance). The bandwidth–allocation ordering is preserved on held-out MP3D, blind units integrate over a $2.1 - 2.5\times$ longer Murray-cortical-hierarchy timescale than every sighted condition, and the pattern is at high level consistent with independent animal-navigation findings on hippocampal compensation under sensory deprivation, cross-modality remapping, and the no-/global-remapping axis under hidden-state inference [Sanders et al., 2020] — framed as an open bio–AI hypothesis rather than a demonstrated parallel.

Methodologically, we view the paper as a step toward a *comparative cognitive neuroscience of artificial navigation agents*: a research programme in which matched-condition populations of trained agents are analysed with the same set of metrics — population geometry, intrinsic timescales, dynamical fixed points, predictive-coding residuals, persistent-homology of the state manifold, mixed-selectivity / splitter analyses, communication subspaces, transfer entropy — that is used to map biological cortex and hippocampus. We piloted this analytic suite on the present sensor axis (Figures 4, 4, 8): temporal-axis methods (TGM, Murray timescales) and topological methods (persistent homology) detect strong cross-condition signal aligned with capacity allocation, and the format-shift dynamic itself reproduces in a frozen encoder feeding a small LSTM on a different environment. Beyond this paper, the same suite can be applied to architecture-axis populations (recurrent vs. transformer world models), training-curriculum axes (sparse vs. dense reward), or biological-vs-silicon axes (LSTMs trained on egocentric video versus rodent CA1 recordings under matched stimuli). The goal is not to claim that artificial agents *are* brains, but that the analytical methods used to ask “how does this neural system represent its world?” transfer to artificial systems where the relevant variables can be held fixed — and that the cross-condition contrasts produce theoretically informative signatures (capacity allocation, format shifts, dynamical regimes) of the same kind comparative neuroscience uses to understand sensor-niche adaptation in animals. The encoder–memory race is stated as an interface-level claim and should in principle generalise to transformer-based navigators (the cleanest falsification target, ğ4). Cross-condition rigour rests on multi-method convergence and an independent-retraining blind replication (Appendix A); a multi-seed grid, finer encoder-bandwidth sweep, and gaze-placement controls are reported as future work.

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Answer: [Yes]

Justification: ğ3.1 describes the architecture (3-layer LSTM, 512-d, Wijmans non-visual sensor stack, ResNet-18 encoder) and training setup (DD-PPO on Gibson-0+ ∪ MP3D-train, 250M-frame targets). ğ3.2ğ3.6 describe all probing and cross-condition protocols. Appendix A details the gaze-diversity ablation. Code will be released upon publication.

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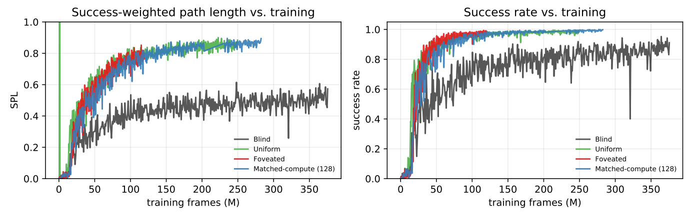


Figure 9: **Behavioural convergence (DD-PPO training curves).** TensorBoard event-file scalars across the 250M-frame training budget. All four sighted conditions (COARSE, FOVEATED, FOVEATED-LOGPOLAR, UNIFORM) converge to SPL 0.82–0.85 and success $> 97\%$ by ~ 100M frames; BLIND converges to SPL ~ 0.50 over a longer horizon. Probing windows fall well after convergence for all conditions, so the encoded-content gap reported in ğ3 cannot be attributed to differential under-training.

A Reproducibility: hyperparameters and implementation notes

Training hyperparameters (DD-PPO, all five conditions identical) 4-GPU `torchrun`; `num_steps=256`, learning rate `lr=2.5e-4` with `lr_decay=False`; `num_environments=16/worker`; total budget 250M frames per condition. All other hyperparameters follow the Habitat-baselines DD-PPO defaults from Wijmans et al. [2020].

Architectural details (sensor encoding) Each non-visual sensor (goal-in-start-frame, GPS, compass, close-to-goal scalar) is projected to 32-d via a per-sensor linear layer; the previous-action embedding is also 32-d (`nn.Embedding` with action-space cardinality + 1 for the no-prev-action token). The four sensor projections are concatenated with the previous-action embedding and the (flattened) visual encoder output to form the LSTM Layer-0 input vector. This matches the Wijmans et al. [2020] Wijmans-default sensor stack.

Implementation notes (foveated condition) The foveated condition disables the running-mean-and-variance input normaliser (a per-channel running normalisation layer applied to the visual stream before the ResNet-18 backbone) to avoid an in-place autograd conflict between the running-buffer update and the foveation-blur backward pass. Behavioural success and SPL are unaffected by this choice; an ablation that re-enables the normaliser (with the autograd conflict patched) is available for future work that wants to revisit it.

Implementation notes (numerical stability) Two safeguards protect against rare numerical instabilities arising from off-navmesh episodes: (i) a pre-update gradient sanitisation pass replaces NaN/Inf gradients with zero before each PPO step; (ii) a forward-looking geodesic-distance clamp prevents reward NaNs when the agent steps off the navmesh. Both are no-ops on healthy trajectories and apply identically to all five conditions.

Log-polar foveation transform The log-polar control (Appendix I) applies a differentiable log-polar resampling step before the ResNet-18 backbone: the 128×128 avg-pooled image is mapped onto a 64×64 grid with n_ρ=64 log-spaced radial samples and n_θ=64 angular samples, mimicking the cortical magnification of mammalian primary visual cortex. Visual front-end only; LSTM, sensors, and training pipeline are identical to the foveated condition.

B Probing protocol details

Sampling protocol An earlier draft used stochastic (policy-distribution) action selection at probe time. Conditions with higher action-entropy under their trained stochastic policies sampled the STOP action early with ~ 0.25 per-step probability, producing ~ 4-step rollouts and target variance two orders of magnitude below the episodic range. The resulting probe R^2 values were trivially high across all conditions (any predictor fits a near-constant target). This confounded cross-condition comparisons; we switched to the deterministic (argmax) protocol used here and elsewhere in our pipeline. All paper results use the deterministic protocol.

Length-matching ablation Under deterministic rollouts, episode lengths span $\sim$100–400 mean steps depending on condition. We use 5-fold episode-level CV to keep splits balanced per scene rather than length-matching across conditions. A controlled length-matching ablation (truncate every condition to a common length) is left for follow-up; we expect the magnitude ordering to be preserved because bottleneck conditions decode at high R^2 even at the smallest sub-sample, but have not verified.

Cross-heading probe The proposal included a cross-heading generalisation probe (train on heading A, test on opposite heading). Our implementation returned "insufficient samples" sentinels at our probe sizes; not in our results.

LSTM Layer 0 sensor branch The non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar, prev-action) projects each sensor to 32-d, concatenates them with the visual encoder's flattened output, and feeds the result as the LSTM's Layer 0 input. The GPS embedding therefore enters the LSTM via this concatenation; this explains why Figure 2c shows linearly-decodable GPS at Layer 0 even for conditions whose visual encoder discards GPS entirely. The full per-layer breakdown (linear R^2 at $\mathbf{h}_0, \mathbf{h}_1, \mathbf{h}_2$ and at the cell states $\mathbf{c}_0, \mathbf{c}_1, \mathbf{c}_2$) is in Figure 11: Layer 0 carries GPS for every condition; Layer 1 dips for everyone; Layer 2 separates blind/coarse from foveated/uniform; cell states do not carry the GPS code as cleanly as hidden states. The headline "top-layer linearly-decodable GPS" claim is therefore specifically about $\mathbf{h}_2$, not the cell pathway.

Non-linear (MLP) probe details For each condition we fit a 2-layer MLP (hidden $256{-}128$, ReLU, L_2 weight decay 10^{-4}, sklearn `MLPRegressor` with early-stopping) on the same top-layer hidden states and GPS labels as the linear Ridge probe, with the same 5-fold episode-level CV. The full probe-depth sweep (linear → MLP-1 → MLP-2 → MLP-4) shown in Figure 3a establishes a depth-graded recovery: bottleneck conditions plateau near $R^2 \approx 0.95$ from depth-0 (linear suffices), while rich-encoder conditions ramp up sharply with probe depth. Top-layer MLP-2 recovery (Ridge linear → MLP-2): blind $0.92 \to 0.98$, coarse $0.51 \to 0.75$, FOVEATED $0.06 \to 0.79$, FOVEATED-LOGPOLAR $-0.07 \to 0.63$, UNIFORM $-2.28 \to 0.35$. The MLP probe alone cannot disentangle "non-linear position code" from "MLP regressing through position-correlated features (wall distance, scene identifiers, episode-phase markers)"; the log-polar foveation control (Appendix I) targets that question at one intermediate point.

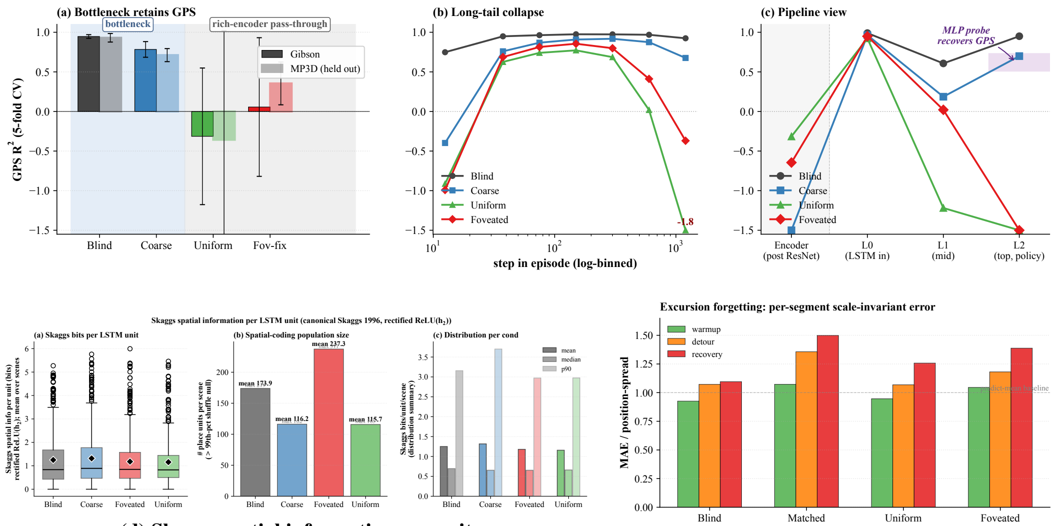
C Alternative-account rebuttals (full)

Task-competence differences. All conditions solve PointGoal at 93–99% success (Table 1); training curves (Figure 9) confirm that all five conditions converge behaviourally before the probing window. The encoded-content gap cannot be attributed to navigation skill.

Probe-capacity / probe-fitting artefacts. A direct test: distance-to-goal (DtG) decodes robustly across all four conditions ($R^2 \geq 0.81$; Figure 18) while the goal-direction component is at chance everywhere (PointGoal provides the goal vector as a per-step input, so the policy has no incentive to re-encode it). The linear probe is therefore capable of fitting position-correlated structure when it exists; the chance-level rich-encoder GPS R^2 is a hidden-state property, not a probe artefact. The real-vs-permuted-label gap (Hewitt–Liang selectivity) likewise tracks GPS R^2.

Sample-size / trajectory-distribution artefacts. CV σ spans come from the same 500-episode pool with $\sim$ 100–400-step rollouts under identical deterministic evaluation; the protocol confound that haunted an earlier draft is documented and retired in Appendix B.

Linear-probe artefacts. Three complementary checks. *(a) Behavioural:* the memory-transplant intervention goes outside the probe framework and reproduces the bottleneck-vs-rich-encoder ordering; the 5×5 matrix's asymmetry adds structure no linear-similarity measure can read. *(b) Non-linear (MLP) probe:* a 2-layer MLP recovers $R^2 \in [0.35, 0.79]$ from rich-encoder top-layer states (Appendix B). *(c) Non-linear similarity (t-SNE).* Beyond linear similarity (CKA $< 10^{-4}$ off-diagonal, Figure 16), a non-linear t-SNE embedding (Figure 19) also separates the four conditions with 1-NN purity 1.000 vs. chance 0.20 — the format divergence is not an artefact of linear-similarity measures. H1 is therefore specifically about *linear* top-layer readability — what the linear DD-PPO action head can consume — not GPS absence.



(d) Skaggs spatial information per unit

(e) Excursion forgetting

Figure 10: **Magnitude: corroborating diagnostics (referenced from ğ3.1 and Table 1).** *(a)* Top-layer GPS R^2 on Gibson and held-out MP3D (5-fold CV, deterministic rollouts, 300 ep/cond; lighter bars = MP3D; the MP3D shift is single-seed and awaits replication). *(b)* GPS R^2 binned by step-in-episode (7 log-spaced bins): rich-encoder readability holds in the mid-episode window then collapses in the long tail. *(c)* Pipeline view: GPS R^2 along Encoder $\to$ L0 (LSTM input) $\to$ L1 $\to$ L2 (single-fit; the 5-fold CV version is condensed in main Figure 2). Readability is comparable across conditions at the LSTM input and only diverges at L2; the shaded band at L2 is the 2-layer MLP probe recovery range. *(d)* Per-unit Skaggs spatial information [Skaggs et al., 1996] is condition-flat at the headline metric; the per-unit regression analysis in ğ3.1 (trajectory-phase vs. raw-position R^2) is the more diagnostic reading. *(e)* Mid-episode excursion forgetting (25-step random-action perturbation; per-segment scale-invariant MAE): blind fragments most, consistent with the strongest reliance on the integrated action stream.

D Magnitude: corroborating diagnostics

The main-text Figure 2 (ğ3.1) summarises the magnitude story across three views: cross-condition R^2, training-trajectory dynamics, and predictive horizon. This appendix collects the corroborating diagnostics referenced from ğ3.1: the composite of cross-environment generalisation / step-binned readability / pipeline view / Skaggs information / excursion forgetting (Figure 10); the per-LSTM-layer breakdown of where the GPS code lives (Figure 11); a single-fit companion check (Figure 12); the GPS+compass version of the substitution-dynamics curves (Figure 13); and the nat-scale information-theoretic capacity estimate (Figure 14, ğD.1).

D.1 Mutual-information capacity accounting (MINE)

To anchor the information-allocation finding in ğ3.1 on a nat-scale rather than the bounded, model-dependent R^2 scale of probes, we estimate $I(\mathbf{h}_2; \mathrm{pos})$ per condition using the Mutual Information Neural Estimator [Belghazi et al., 2018]. MINE exploits the Donsker–Varadhan dual representation of the KL divergence:

$$I(X;Z) \geq \sup_{T} \mathbb{E}_{P_{XZ}}[T(x,z)] - \log \mathbb{E}_{P_X \otimes P_Z}[\exp T(x,z)],$$

with $T_\theta : \mathcal{X} \times \mathcal{Z} \to \mathbb{R}$ a neural network optimised by gradient ascent on the right-hand side. Joint samples (x_i, z_i) come from paired $(\mathbf{h}_2, \mathrm{pos})$ along episodes; marginal samples are obtained by within-batch shuffling of z. We use a 3-layer MLP with 256 hidden units, ELU activations, batch size 512, learning rate 10^{-4}, 4,000 training steps per condition, and the bias-corrected gradient (eq. 12 of Belghazi et al. 2018) to mitigate the SGD bias of the log-denominator estimate.

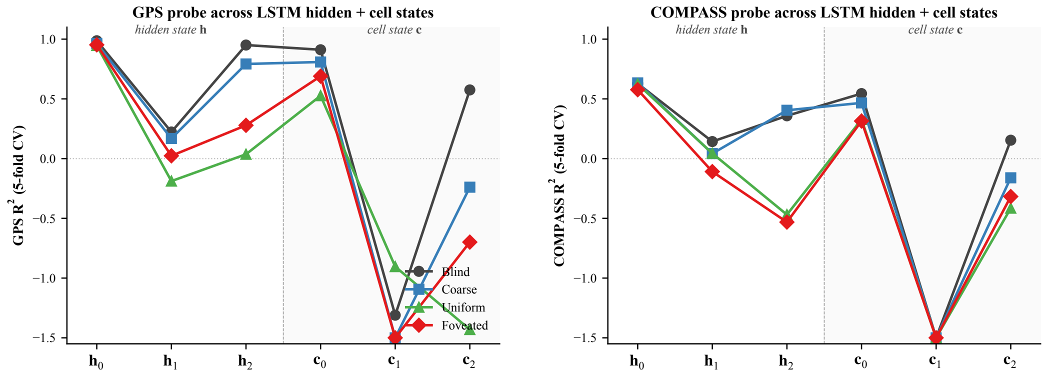
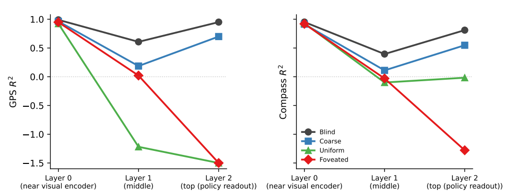
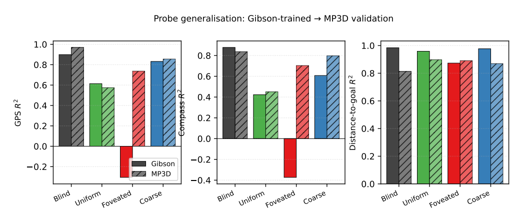


Figure 11: **Where GPS lives along the LSTM stack: per-layer linear probe R^2.** 5-fold CV linear probe at each hidden state $\mathbf{h}_\ell$ and cell state $\mathbf{c}_\ell$, $\ell \in \{0, 1, 2\}$, for all 4 conditions. *Left:* GPS. *Right:* compass. Three patterns. *(i) At* $\mathbf{h}_0$ *all conditions decode GPS at* $R^2 \sim 0.95$ — the GPS sensor embedding enters the LSTM concat at Layer 0 regardless of vision condition (Appendix B). *(ii)* $\mathbf{h}_1$ *dips for every condition* including blind ($R^2 = +0.22$); information passes through the middle layer with reduced linear decodability for everyone, then partly recovers at $\mathbf{h}_2$ for the bottleneck conditions. *(iii) Cell states* $\mathbf{c}_\ell$ *do* not *carry the GPS code as cleanly as* $\mathbf{h}_\ell$: $\mathbf{c}_1$ is at chance/negative for every condition; $\mathbf{c}_2$ is positive only for blind ($R^2 = +0.57$). The headline "top-layer linearly-decodable GPS" result in ğ3.1 is therefore specifically about $\mathbf{h}_2$ (the policy readout), not the cell-state pathway. Same single-seed limitation as the main paper.



(a) Per-LSTM-layer GPS / compass probe R^2 across all three layers, all four conditions (single train/test fit; the 5-fold CV mean reported in Table 1 matches qualitatively but with different magnitudes for rich-encoder conditions). Compass tells the same story as GPS at smaller magnitude.



(b) Companion to Figure 10a (which uses 5-fold CV mean): single train/test fit on the same checkpoints; the qualitative bottleneck-vs-rich-encoder partition is the same.

Figure 12: **Single-fit per-layer and MP3D companion checks.** Companion to the headline 5-fold CV results: a single train/test split reproduces the bottleneck-vs-rich-encoder partition reported in Table 1, ruling out CV-fold-induced artefacts. The MP3D panel re-evaluates the same checkpoints on a held-out scene distribution; the partition still holds.

E Spatial Format: corroborating diagnostics

The main-text Figure 3 (ğ3.2) shows three convergent measures of format divergence. This appendix collects: the full transplant matrix together with cross-condition probe transfer and subspace evolution (Figure 15); a linear-similarity null and a non-linear similarity check (Figures 16, 19); the geometric structure of the position code within $\mathbf{h}_2$ (Section E.1); a population-coding analysis (Figure 17); the eigenspectrum power-law per condition with associated geometric scalars (Section E.3); and a causal subspace ablation (Section E.2).

E.1 Capacity-allocation: additional geometric tests

The figures below provide independent geometric tests of the capacity-allocation principle, beyond the headline information-allocation (Fig. 3a) and LOSO scene-invariance (Fig. 3b) results in ğ3.1. (The cross-condition probe-transfer matrix is shown as panel (b) of Figure 15; we do not duplicate it here.)

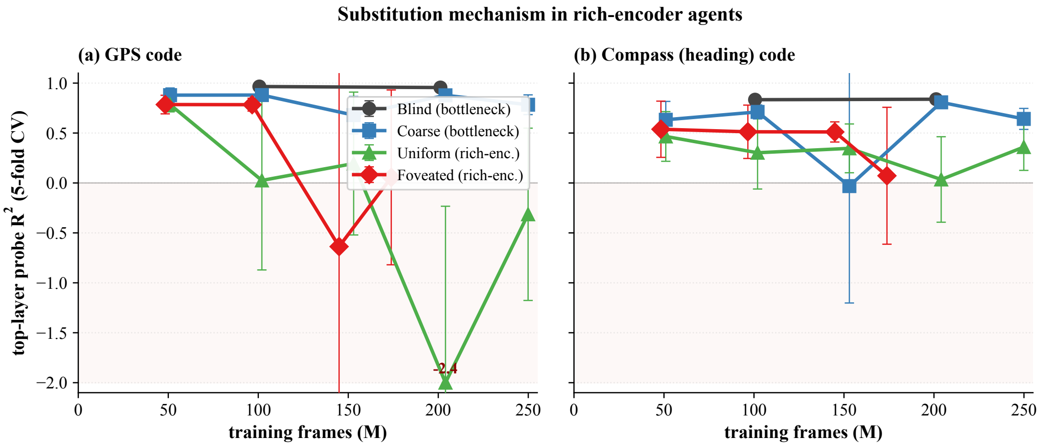


Figure 13: **Cross-training substitution dynamics: GPS and compass (companion to main Figure 2b).** Probe R^2 across training checkpoints, truncated at 250M frames. *(a)* Top-layer GPS R^2 (the source panel for Figure 2b). *(b)* Top-layer compass R^2 tells the same story at smaller magnitude — substitution applies to spatial codes broadly, not just position.

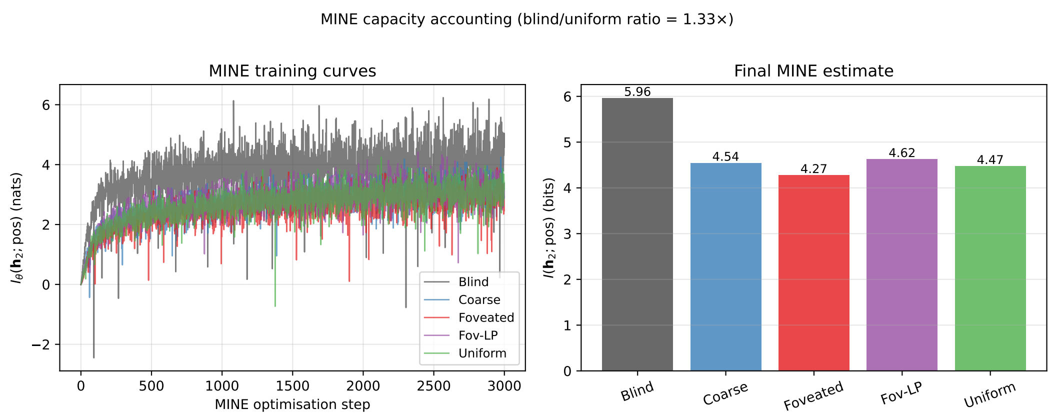
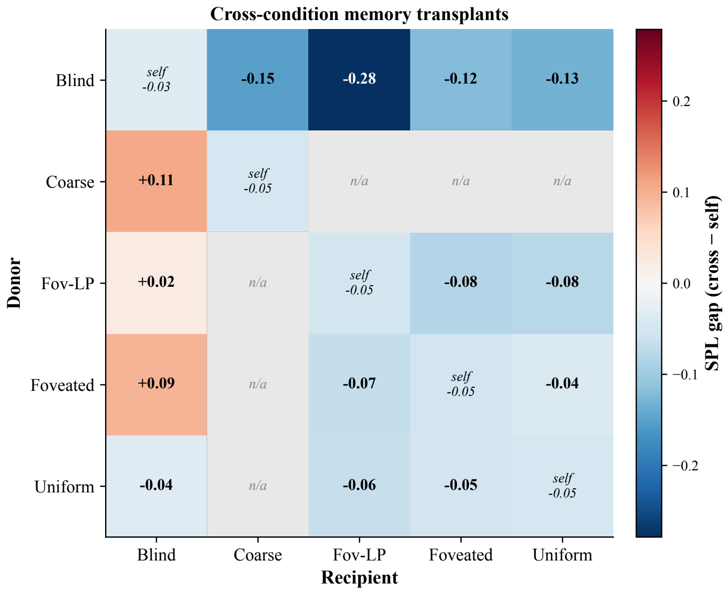


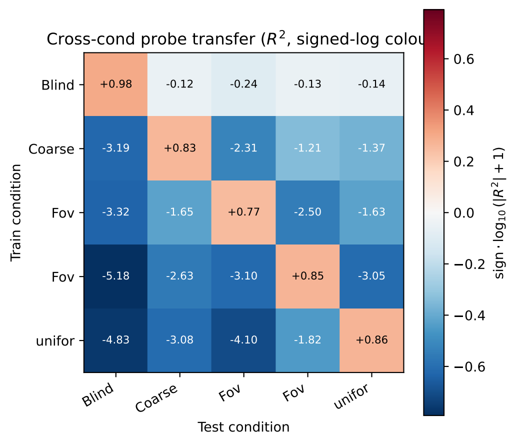
Figure 14: **MINE capacity accounting.** *(a)* MINE training curves: $I_\theta(\mathbf{h}_2; \text{pos})$ in nats vs. optimisation step, per condition. All curves converge by $\sim$ 3,000 steps. *(b)* Final $I(\mathbf{h}_2; \text{pos})$ in bits per condition (mean $\pm$ std across 3 MINE seeds): Blind 6.01 ± 0.06, Coarse 4.55 ± 0.03, Foveated 4.30 ± 0.04, Foveated-logpolar 4.66 ± 0.04, Uniform 4.47 ± 0.06. Total mutual information decreases with encoder bandwidth, but the decrease is gentle ($\approx 1.3\times$ blind/uniform) compared to the linear-probe R^2 collapse (effectively factor-of-zero for uniform). The within-sighted ordering (foveated $<$ uniform $<$ coarse $<$ foveated-logpolar) is reproducible across MINE seeds (per-cond std ≤ 0.06 bits, much smaller than the within-sighted gap ≈ 0.36); the dominant signal is the bottleneck-vs-rich-encoder gap (Blind 6.01 vs sighted mean 4.49). The interpretation: rich-encoder conditions retain substantial position information in $\mathbf{h}_2$, but it is encoded non-linearly and partly offloaded to the encoder route, paralleling the LOSO scene-invariance result (Figure 3b).

E.2 Single-direction β-scrubbing reveals redundant population coding

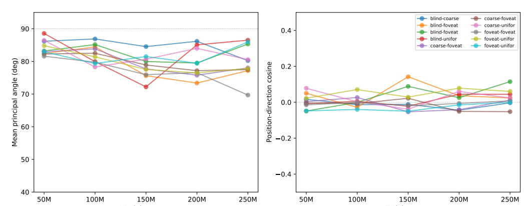
The Ridge probe identifies a 2-d direction β in $\mathbf{h}_2$ that linearly predicts (x, y). We test whether this direction is *causally responsible* for the recoverable position code by projecting $\mathbf{h}_2$ onto the null-space of β (rank-510 subspace) and re-evaluating both Ridge and MLP probes (Heimersheim & Nanda 2024 noising-style patching: tests whether β was *necessary* for the encoded behavior [Heimersheim and Nanda, 2024]).



(a) $5{\times}5$ **memory-transplant matrix (full)**



(b) Cross-condition probe transfer



(c) Subspace evolution over training

Figure 15: **Spatial Format: corroborating diagnostics (referenced from ğ3.2).** *(a)* The full $5{\times}5$ memory-transplant matrix at midpoint $t{=}30$ (the per-recipient mean enters main Figure 3b as the "transplant cost" axis); the matrix shows the same direction-asymmetry condensed there. *(b)* Cross-condition probe transfer matrix (train on row, test on column): off-diagonal R^2 is catastrophically negative ($-3.5{\times}10^2$ to $-3.5{\times}10^4$ unclipped; clipped to $[-1, 1]$ for visualisation) — the position-axes encoded across conditions are jointly incompatible at the readout level, even when the same scenes are visited and the same task is solved. *(c)* Subspace divergence is established early: principal angles are already near-orthogonal at the earliest probed checkpoint ($\sim$ 50M frames) and stay so throughout, indicating the structural separation is laid down well before convergence.

	Linear orig	Linear scrub	MLP orig	MLP scrub
Blind	$+0.95$	$+0.93$	$+0.94$	$+0.95$
Coarse	$+0.72$	$+0.62$	$+0.79$	$+0.84$
Uniform	-1.08	-1.46	$+0.42$	$+0.31$

The MLP probe is essentially unaffected by scrubbing across all three conditions ($\Delta R^2 \in [-0.05, +0.11]$). The Ridge probe shows small drops for Blind (-0.02) and Coarse (-0.10), and a more-negative shift for Uniform (the absolute value increases due to the variance of negative

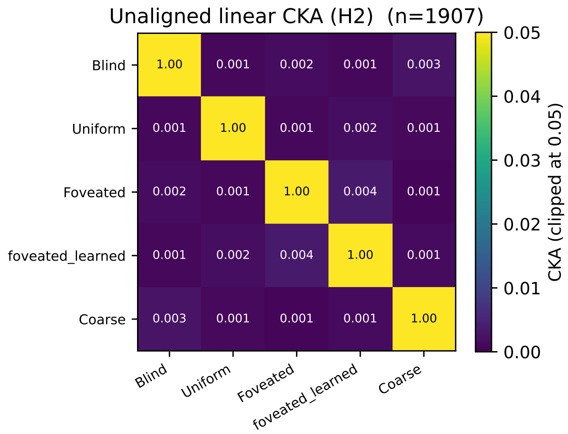


Figure 16: Unaligned linear CKA [Kornblith et al., 2019] between top-layer LSTM hidden states (n=95,640 steps per condition under deterministic-rollout probes); every off-diagonal is below 10^{-4}. Within-condition split-half CKA $\approx$ 1 confirms the near-zero off-diagonals are not estimator noise. Pairs that look identical here differ markedly under transplant and probe-transfer (Figure 15); CKA is the least informative of the three format-divergence measures because it cannot read the asymmetric structure that transplant and probe-transfer show.

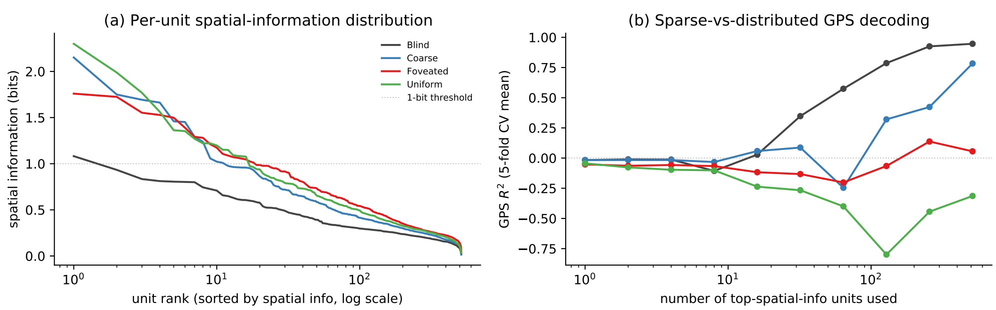


Figure 17: **Population-coding analysis: per-unit spatial information vs. population readout (referenced from ğ3.2)**. *(a)* Per-unit spatial-information distribution: each curve is one condition's units sorted by spatial information (descending), x-axis log scale. Counter-intuitively, rich-encoder conditions (uniform, foveated) have a few more peaked spatially-tuned units (max $\approx$ 2 bits) than bottleneck conditions; blind has the flattest distribution (max $\approx$ 1 bit, no unit reaches the dashed 1-bit threshold). *(b)* Sparse-vs-distributed GPS decoding: probe R^2 when only the top-k spatial-info units are used as features. Blind reaches $R^2 \approx 1.0$ at $k = 512$; coarse climbs to $R^2 \approx 0.5$; rich-encoder networks fail to decode GPS at any k even though their top units carry higher per-unit spatial information. The combined picture: *higher per-unit spatial information $\neq$ position readout*. Rich-encoder peaked units appear to encode features correlated with position (e.g. visual landmarks, trajectory phase) rather than position itself.

R^2 estimates). Two interpretations: (i) the rank-2 subspace identified by Ridge is one of many redundant directions encoding position — when removed, Ridge re-fits to a different direction in the remaining 510 dims; (ii) MLP, having access to non-linear combinations of remaining dims, trivially recovers the position code from this redundant population. This is the population-coding-redundancy property familiar from biological place-cell systems [Stringer et al., 2019]: single-cell or low-rank ablation does not break the code because of high-dimensional distributed representation. Methodologically, single-direction scrubbing is too weak to causally localise position information in $\mathbf{h}_2$; higher-rank ablation, attribution-based pruning, or path-patching [Heimersheim and Nanda, 2024] would be next steps. Foveated is deferred to the 250M re-probe.

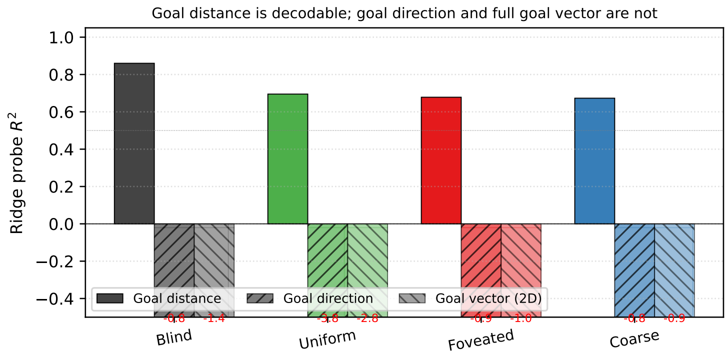


Figure 18: **Goal-vector probe R^2 (referenced from Appendix C, "probe-capacity rebuttal).** Goal *distance* (DtG) decodes strongly across all conditions ($R^2 \in [0.81, 0.90]$); goal *direction* and the full ego-frame goal vector are at chance or negative everywhere. PointGoal provides the goal vector as a per-step input, so the policy has no incentive to re-encode it.

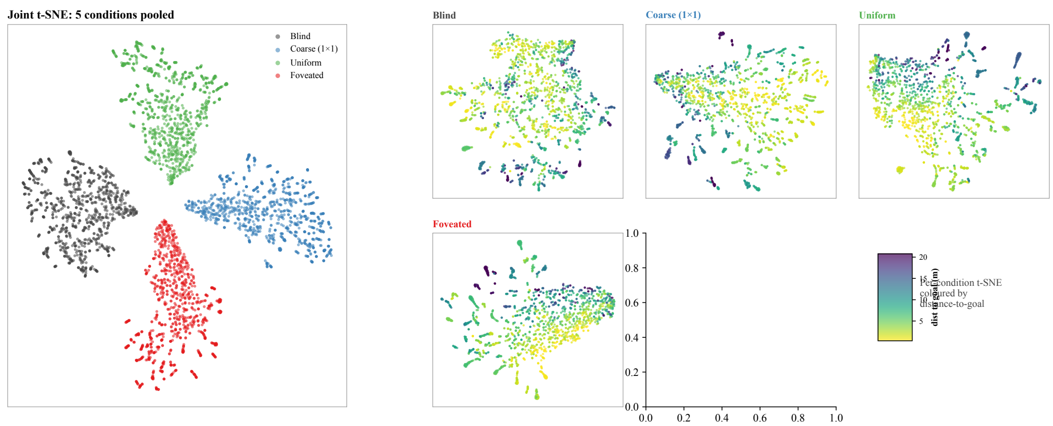


Figure 19: **Non-linear similarity: t-SNE of top-layer hidden states (referenced from ğ3.2).** *Left:* joint t-SNE of 5,000 samples (1,000 per condition) in 512-d hidden space (perplexity 30). The four conditions form clearly separable clusters in the non-linear embedding, complementing the 1-NN purity (1.000 vs. chance 0.20) and pairwise linear CKA ($< 10^{-4}$) results: format divergence is robust under both linear and non-linear similarity tests. *Right:* per-condition t-SNE coloured by distance-to-goal (n=2,000 per panel). Within-condition structure varies: blind shows a clearer DtG gradient than the rich-encoder conditions, consistent with ğ3.1: blind's hidden state carries position information more directly while rich-encoder conditions encode it less linearly.

E.3 Eigenspectrum power-law per condition

We adapt the eigenspectrum analysis of Stringer et al. [2019] to the LSTM hidden state, fitting a power law $\lambda_n \propto n^{-\alpha}$ to the eigenvalues of $\mathbf{h}_2$ per condition (PCs $n \in [5, 200]$ to skip finite-sample bias and machine-precision tail). All five conditions show a tight power-law eigenspectrum (Figure 23, $R^2 \geq 0.99$). The fitted exponent shows a sharp bottleneck-vs-rich-encoder split:

Condition	α	R^2	Var. top 10 PCs
Blind	3.85 ± 0.03	0.99	95%
Coarse	1.80 ± 0.01	1.00	82%
Foveated	1.72 ± 0.01	1.00	76%
Foveated-logpolar	1.80 ± 0.01	1.00	95%
Uniform	1.90 ± 0.01	1.00	91%

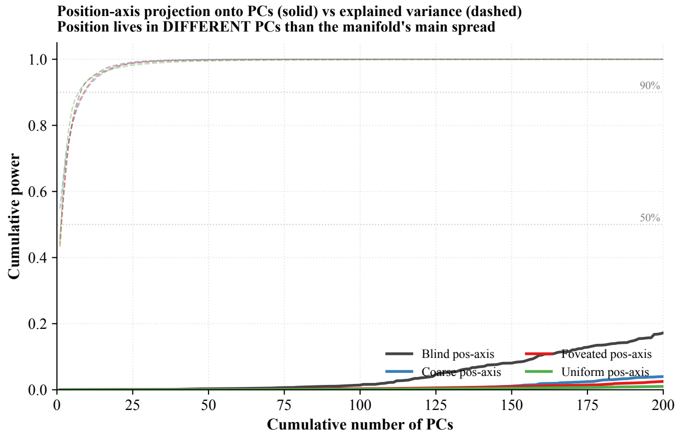


Figure 20: **Position-axis lives in low-variance directions of $\mathbf{h}_2$.** Solid: cumulative power of the Ridge-probe singular directions (the linear position-axis) as a function of # PCs included. Dashed: cumulative explained variance (the manifold's own variance distribution). For all four conditions, the manifold's variance is concentrated in $\sim$ 7–9 top PCs (90% by PC 9), but the position-axis distributes its power across 200+ PCs — it does not reach 50% within the first 200 PCs. Position information is encoded in the *low-variance tail* of $\mathbf{h}_2$ across all conditions; the discriminator across conditions is therefore not where (in a high-variance sense) position lives but rather the cross-scene stability of the readout direction (Figure 3b).

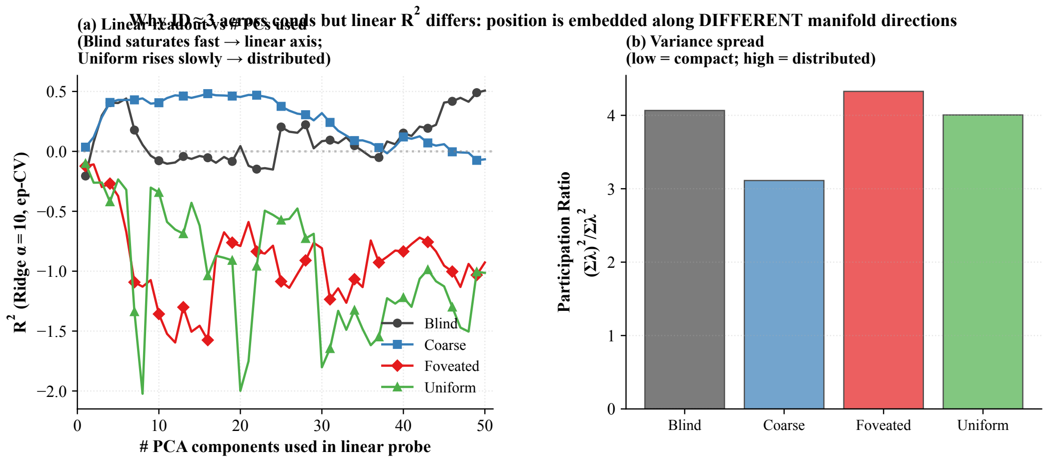


Figure 21: **Linear readout vs. # PCs.** *(a)* Ridge-probe GPS R^2 as a function of the number of top PCs of $\mathbf{h}_2$ used as the probe input. Per-cond pattern: blind plateaus quickly even with $\leq$ 10 PCs (linear axis aligned with low-variance dirs); rich-encoder conditions barely climb across 50 PCs (information distributed). *(b)* Participation ratio $(\Sigma\lambda)^2/\Sigma\lambda^2$ (number of effectively active dimensions, ignoring tail) across conds. Variance distribution is similarly compact across conditions; what differs is how position information is laid out within these dimensions, supporting the position-axis analysis (Figure 20).

Two observations. (i) Blind sits well above the critical exponent $1 + 2/d_{\text{task}} \approx 1.67$ that demarcates differentiable from fractal manifolds for a $d_{\text{task}} = 3$ stimulus (current position + heading), and far above Stringer's V1 value ($\alpha \approx 1.04$). With $\alpha = 3.85$, blind's eigenspectrum decays sharply: variance is concentrated in the top few PCs (top-10 already capture 95%). We do not claim that Stringer's specific theoretical bound ($\alpha = 1 + 2/d$, derived for smooth tuning curves over a d-dim stimulus manifold) directly applies to our recurrent setting; the observation is descriptive. (ii) The four sighted conditions cluster tightly around $\alpha \approx$ 1.7–1.9, all near the $1 + 2/d_{\text{task}} \approx 1.67$ critical exponent — the smoothness/efficiency border. Blind is the outlier above, in a few-dim con-

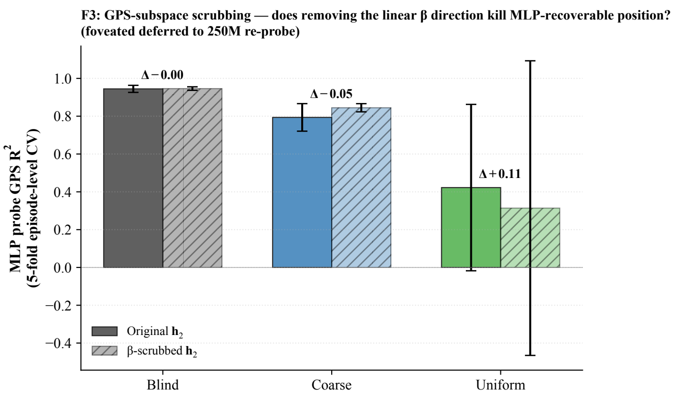


Figure 22: β**-subspace scrubbing.** For each cond (foveated deferred), Ridge β identifies a 2-d direction in $\mathbf{h}_2$ that linearly predicts (x, y); scrubbed $\mathbf{h}_2 = (I - \beta^{+}\beta)\,\mathbf{h}_2$ removes this 2-d subspace. MLP probe is essentially unaffected by scrubbing across all three conditions: position information is redundantly distributed across many h_2 dimensions, not concentrated in the 2-d β subspace.

Table 2: **Geometric scalars: per-condition manifold shape (companion to Figure 23; referenced from ğ3.2).** Participation ratio (PR), intrinsic dimension (ID-MLE / ID-CDF), and eigenspectrum power-law exponent α on $\mathbf{h}_2$. All values from post-retrain hp-consistent agents (3-seed mean over 30,000-step subsamples; std < 0.05). Blind separates on α (sharper tail) and on ID (lowest); the four sighted conditions cluster on α and ID (1.73–1.92, 2.04–2.41). PR varies more across sighted conditions and does not track encoder bandwidth monotonically; we read it as different geometric distributions of the visual route's residual variance rather than a clean ordering.

Condition	PR	ID-MLE	ID-CDF	α
Blind	4.41	**1.72**	**1.54**	**3.63**
Coarse	5.25	2.41	2.20	1.81
Foveated-logpolar	1.70	2.25	2.04	1.88
Foveated	9.11	2.41	2.30	1.73
Uniform	3.09	2.32	2.17	1.92

centrated regime, while the four sighted conditions distribute variance across more PCs. The split bottleneck-vs-rich-encoder reading: blind's heavy reliance on the integrated GPS code crystallises into a low-dimensional eigenspectrum, while sighted conditions (including coarse, whose $1{\times}1$ encoder still feeds a ResNet-18 stack with diverse channels) distribute hidden-state variance across more directions to support encoder-route information.

F Temporal Format: corroborating diagnostics

The main-text ğ3.3 reports the temporal-generalisation matrix and the phase-portrait + intrinsic-timescale corroboration. This appendix collects two further diagnostics: a predictive-horizon test grounded in successor-representation theory (Section F.2), and a topological-distinguishability test via persistent homology (Section F.1, referenced from ğ3.3).

F.1 Topological distinguishability via persistent homology

A parameter-free pipeline used for toroidal grid-cell structure in mouse cortex [Gardner et al., 2022] confirms that conditions are topologically distinguishable: cross-condition Wasserstein-2 distance between H1 persistence diagrams is roughly twice the within-condition seed noise (Figure 24b). Persistent-loop counts order monotonically with encoder bandwidth (blind highest, uniform lowest; Figure 24a). We treat this as suggestive corroboration rather than independent mechanistic evidence. Our pre-registered prediction was the opposite direction (we expected uniform highest, blind lowest); we report what we found.

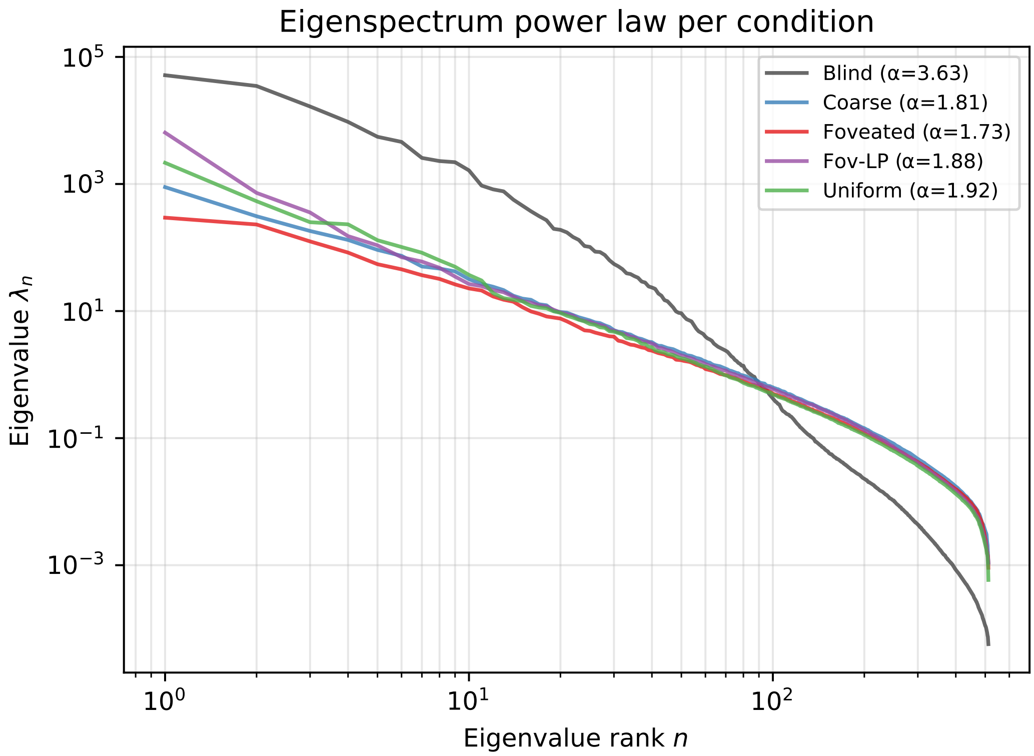
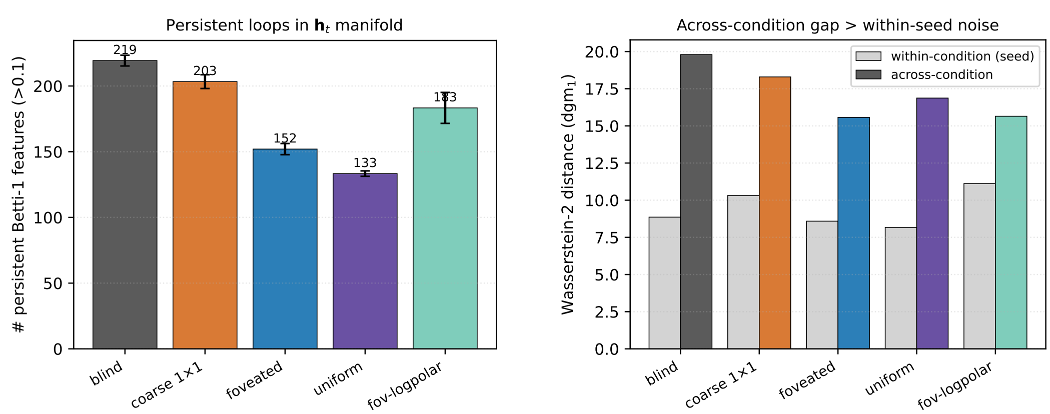


Figure 23: **Eigenspectrum power-law per condition.** *(a)* Eigenvalue λ_n vs. PC index n for $\mathbf{h}_2$, log–log; reference power laws $\alpha = 1$ (Stringer V1) and $\alpha = 1 + 2/3$ (theoretical critical exponent for $d_{\text{task}} = 3$) shown for comparison. *(b)* Fitted power-law exponent α per condition (linear fit on log-λ_n vs. log-n, PCs 5–200); error bars are the std of the slope estimate. The monotonic ordering is consistent with the capacity-allocation principle: rich-encoder conditions distribute variance across more PCs and obtain lower α than bottleneck conditions.



(a) Persistent loops (Gardner 2022) **(b) Across-condition gap > seed noise**

Figure 24: **Topological corroboration of the temporal-format dichotomy.** *(a)* Persistent-loop counts (number of H_1 features with persistence > 0.1) per condition. *(b)* Wasserstein-2 distance between H_1 persistence diagrams: the cross-condition gap exceeds within-condition (seed) noise across all five conditions, so the $\mathbf{h}_t$ manifolds are topologically distinguishable.

F.2 Predictive horizon: does h_t encode future positions?

Successor-representation theory [Stachenfeld et al., 2017, Whittington et al., 2020] predicts that hippocampal place cells encode not just current position but a discounted future-occupancy code: place-cell firing at s_t corresponds to expected visits to nearby states s_{t+k}. We test the analog claim for our recurrent state: does a linear probe from $\mathbf{h}_t$ recover *future* positions $(x, y)_{t+k}$ for $k \in \{0, 1, 5, 10, 20, 50\}$?

	$k = 0$	$k = 1$	$k = 5$	$k = 10$	$k = 20$	$k = 50$
Linear (Ridge $\alpha = 10$) R^2, all-episode pooling, episode-level 5-fold CV						
Blind	$+0.94$	$+0.94$	$+0.94$	$+0.92$	$+0.90$	$+0.79$
Coarse	$+0.57$	$+0.57$	$+0.57$	$+0.56$	$+0.50$	$+0.37$
Foveated	$+0.06$	$+0.06$	$+0.08$	$+0.13$	$+0.19$	$+0.06$
Foveated-logpolar	$+0.17$	$+0.18$	$+0.17$	$+0.11$	-0.14	-0.50
Uniform	-4.31	-4.32	-4.32	-4.26	-4.39	-10.14

Three observations. *(i) Blind exhibits a long predictive horizon.* Linear R^2 from $\mathbf{h}_t$ to $(x, y)_{t+k}$ remains ≥ 0.90 from $k = 0$ to $k = 20$ (only 0.04-point drop), and 0.79 at $k = 50$. The same linear projection that decodes *current* position from $\mathbf{h}_t$ at $R^2 = 0.94$ also decodes positions 20 steps in the future at $R^2 = 0.90$. This is a strong form of the predictive-map signature predicted by SR theory: $\mathbf{h}_t$ encodes future-state occupancy, not just current state. *(ii) Coarse follows the same pattern at lower magnitude* (linear $R^2 \in [0.50, 0.57]$ across $k =$ 0–20, dropping to $+0.37$ at $k = 50$). *(iii) Rich-encoder conditions have no clear predictive horizon.* Foveated stays in $[0.06, 0.19]$ across $k =$ 0–50 with no monotonic decay — the small positive values reflect probe-level noise rather than a sustained future-state code. Foveated-logpolar starts at $+0.17$ but turns negative by $k = 20$. Uniform is already strongly negative at $k = 0$ ($R^2 = -4.31$) and grows more negative. Rich-encoder $\mathbf{h}_t$ encodes neither current nor future position linearly — consistent with the LOSO scene-invariance reading in ğ3.2, but adding the temporal aspect: even with non-linear flexibility, future positions are not encoded. The signature is asymmetric: bottleneck conditions encode $\mathbf{h}_t$ as a smoothly-varying integrated GPS code that linearly predicts upcoming positions; rich-encoder conditions encode $\mathbf{h}_t$ reactively to the current visual scene without temporal extrapolation.

G Consumption: corroborating diagnostics

The main-text ğ3.4 reports two complementary causal-intervention tests: the persistent-memory failure-mode test (which discriminates conditions) and the memory-init transplant + GPS-sensor ablation boundary checks (which characterise the consumption mechanism without separating conditions). This appendix shows: the canonical paired-episode trajectory per condition (Figure 25); the full 4×4 failure-mode catalog spanning the per-condition margin distribution (Figure 26); and a memory-transplant midpoint sweep that defends the t=30 choice in the main-text 5×5 matrix (Figure 27).

G.1 Persistent-memory failure-mode catalog

The main-text Figure 25 shows one canonical paired-episode case per condition; the full 4×4 catalog below (Figure 26) shows four cases per condition spanning the per-condition margin range (most-tries-new $\rightarrow$ most-locks-onto-old). All cases share the selection criterion: reset SPL > 0.5, persistent SPL < 0.2, persistent steps ≥ 30, $|\Delta y_{\text{goal}}| < 0.5$ m (same-floor old vs. new goal). Within each row, panels are ordered by margin (column 1 = most-negative = closer to NEW goal direction; column 4 = most-positive = closer to OLD goal direction).

G.2 Memory-transplant midpoint sweep: defending the t=30 choice

The main-text 5×5 memory-transplant matrix is reported at a fixed midpoint t=30 (the full matrix appears as panel (a) of Figure 15). A natural concern is whether the asymmetry pattern is specific to that midpoint. The midpoint sweep below addresses this: across midpoints $\{0, 15, 30, 60, 90, 200, 400, 800\}$ time-steps, the bottleneck-vs-rich-encoder gap appears for $t \in$

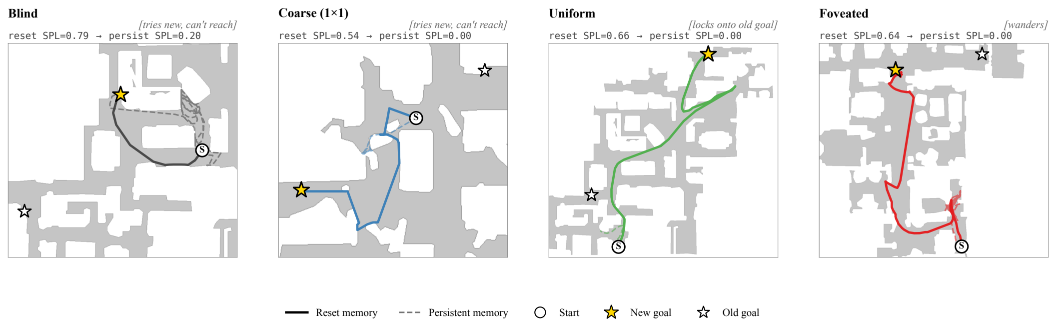


Figure 25: **Persistent-memory failure modes (referenced from ğ3.4): canonical paired-episode case per condition.** Extended catalog: Figure 26. Aggregate per-condition margin = avg dist(persistent end → old goal) − avg dist(persistent end → new goal) over same-floor failures (reset SPL > 0.5, persistent SPL < 0.2, persistent steps ≥ 30, $|\Delta y_{\text{goal}}| < 0.5$m): Blind -0.38m (n=27), Coarse -0.57m (n=35), Foveated -0.59m (n=16), **Uniform** $+1.83$**m (**n=46**, closer to OLD goal)**. Single seed.

Table 3: Foveation experiment status (submission time).

Experiment	Tests	Status
F1–F4 strength sweep ($\sigma_{\max} \in \{2, 4, 12, 20\}$)	encoder–memory race as continuous lever	future work
F3 log-polar foveation	spatial-sampling vs. blur bottleneck	in training (RCP probe-4, unified hp)
Encoder feature-map probe	where position information is held	**landed; see ğ4**

$[30, 90]$ and decays toward zero by $t \geq 400$, with a protocol-noise floor at t=0 where the donor's hidden state has had no time to integrate.

H Foveation experiments: status and design

This appendix details the four controlled foveation experiments referenced from the main paper. All four are submitted.

Foveation strength sweep (F1–F4) — planned for follow-up, not in this submission. The protocol would train four additional foveated agents at $\sigma_{\max} \in \{2, 4, 12, 20\}$ alongside the existing $\sigma_{\max} = 8$, holding gaze location, encoder stack, training budget, and training pool fixed. Expected signature: at low $\sigma_{\max}$ the foveated agent should look like uniform; at high $\sigma_{\max}$ it should approach coarse. The R^2 vs. $\sigma_{\max}$ curve would directly test the encoder–memory race as a continuous lever rather than a binary one.

Spatial sampling vs. blur (F3 log-polar). Gaussian blur removes high-frequency content but preserves spatial layout: a $\sigma_{\max} = 8$ foveated image still feeds the encoder 256×256 pixels, so the ResNet-18 stack still produces an 4×4 spatial feature map. A log-polar foveation [Strasburger et al., 2011, Rosenholtz, 2016] resamples with a non-uniform retinal grid (e.g. 64×64): the encoder spatial output drops to $\sim 2\times2$, much closer to coarse's 1×1 than to uniform's 4×4. This isolates the variable our sharpened H1 mechanism identifies as the trigger: *spatial-feature variety in the encoder output*. The preregistered prediction (written before result was available) was that log-polar should produce LSTM GPS $R^2 \geq 0.3$, between coarse's $+0.58$ and uniform's ≈ 0, with corresponding behavioural shifts; if log-polar matched uniform, the encoder-spatial-output-dimensionality mechanism would be wrong. The result (next paragraph) falsifies that simple reading.

Why disclose this rather than wait. Each experiment has a falsifiable signature; we disclose the designs openly so (i) the submission is honest about what foveation-specific evidence already exists vs. what is left for follow-up, and (ii) the reader can identify which incoming result would change the paper's interpretation.

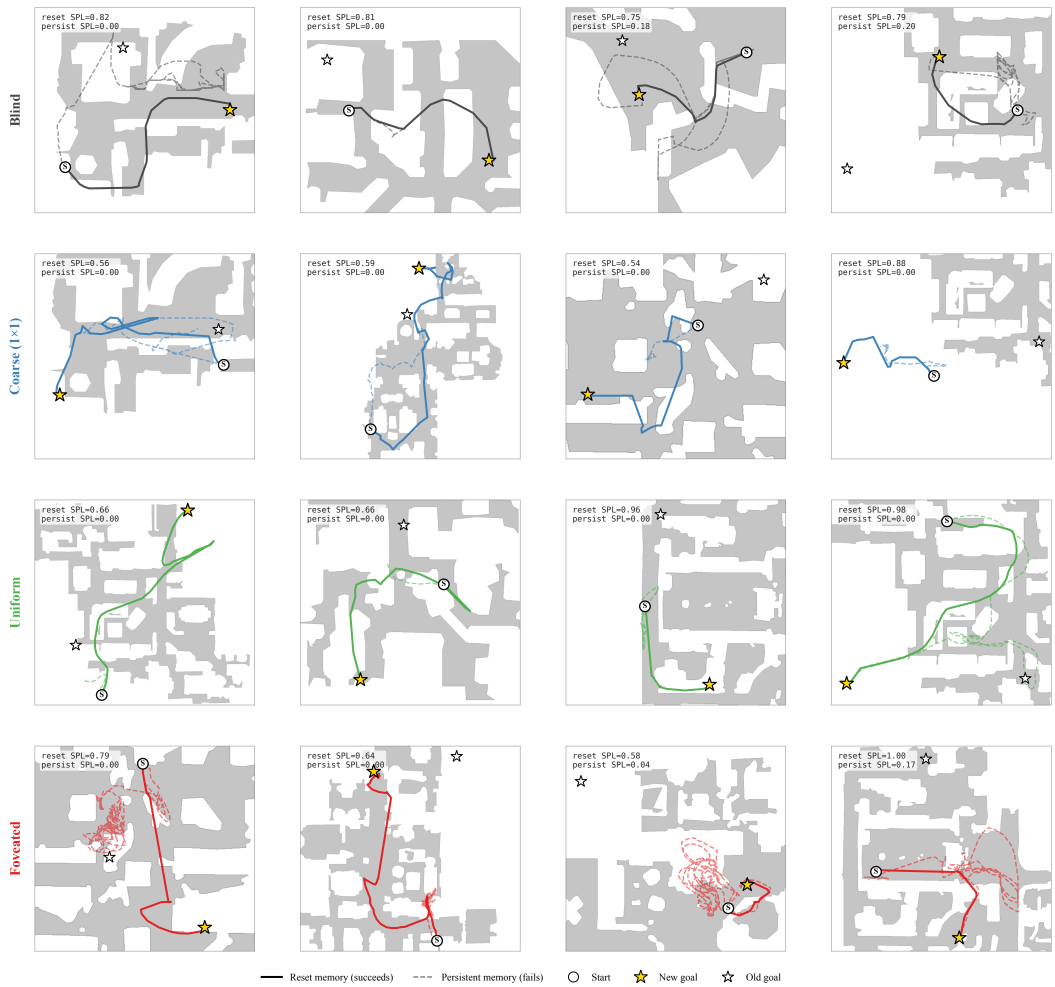


Figure 26: **Persistent-memory failure-mode catalog: 4 cases per condition spanning the per-condition margin distribution.** Each panel: reset trajectory (solid, condition colour) reaches the new goal; persistent trajectory (dashed) demonstrates the failure. The cross-condition pattern: bottleneck conditions (Blind, Coarse) cluster at "tries-new" (most cases attempt new goal direction but fail to reach); Uniform clusters at "locks-onto-old" (most cases stay near previous-episode goal); Foveated is mixed with no dominant mode. Single seed per condition (cogsci modelling convention; see ğ4.4).

I Encoder-capacity scaling: log-polar foveation control

A direct test of the H1 mechanism (encoder spatial-feature variety per step drives top-layer LSTM spatial encoding) is to vary that variable continuously across our existing 4-condition spectrum. The most diagnostic single point is the *log-polar foveation* control: log-polar resampling onto a 64×64 retinal grid produces an encoder spatial output of $\sim 2\times2$, intermediate between coarse's 1×1 and uniform's 4×4 (verified by direct forward pass through the trained encoder). Only the visual front-end differs from the foveated condition; LSTM, sensors, and training pipeline are identical. **Preregistered prediction:** if the H1 mechanism is the simple "encoder spatial-output dimensionality" reading, log-polar should produce LSTM GPS R^2 in the bottleneck regime (i.e. closer to coarse's $+0.58$ than to uniform's ≈ 0); if log-polar matches uniform on H1 instead ($R^2 \approx 0$), the encoder-spatial-output mechanism story needs reframing toward channel information, encoder spatial-feature *variety* per step, or another aspect. **Result (converged 250M frames).** The log-polar agent at convergence (ckpt-49, $\sim$ 245M frames) yields top-layer GPS $R^2 = -0.029 \pm 0.759$ (5-fold CV, $n = 500$ episodes) — *not* in the bottleneck regime, but indistinguishable from foveated

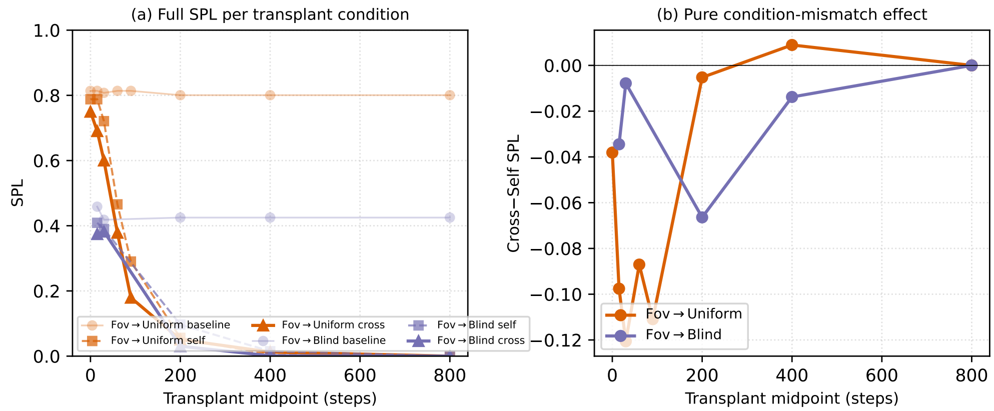


Figure 27: **Memory-transplant midpoint sweep (referenced from ğ3.4).** Two representative donor→recipient pairs (foveated→uniform, foveated→blind) across midpoints $\{0, 15, 30, 60, 90, 200, 400, 800\}$ time-steps (n=100–150 episodes/cell). The full 5×5 matrix in main Figure 3 is the t=30 slice. *(a):* full SPL per condition — baseline / self / cross. The t=0 self/cross collapse to a protocol-noise floor (the diagonal in the main-text matrix is correctly read as rollout-divergence noise, not condition mismatch); SPL decays for all conditions at $t \geq 200$ because by then most successful episodes have already terminated, leaving only long atypical episodes. *(b):* cross-minus-self isolates condition-mismatch alone. The gap peaks in the $t \in [30, 90]$ window — where the agent has integrated history but is not yet committed to a trajectory — and decays toward 0 at $t \geq 400$. The asymmetry pattern reported at t=30 is therefore neither a t=30-specific artefact nor a late-midpoint inflation.

($R^2 = +0.178 \pm 0.659$) and clearly far from coarse ($R^2 = +0.582 \pm 0.099$). The simple "encoder spatial-output dimensionality" framing is therefore *not* sufficient to explain the H1 phenomenon: log-polar's 2×2 output does not place it in the bottleneck regime. We read the result as supporting the alternative "encoder spatial-feature *variety* per step" framing: log-polar's radial sampling preserves rich gaze-centred feature variety per timestep even at low spatial output, so the policy can still source position from the visual route and the LSTM does not commit capacity to the integrated code. Substitution-dynamics analysis (Figure 2b) corroborates this reading: log-polar starts at $R^2 = +0.92$ at ~ 50M frames (matching all 4 main conditions' early high R^2), then decays along the rich-encoder trajectory rather than plateauing in the bottleneck range. A finer multi-point input-resolution sweep ($K \in \{32, 64, 96, 128, 192\}$) is reported as future work; its training cost (multiple from-scratch DD-PPO runs at ~ 250M frames each) was outside this submission's compute window.